

Review

A systematic review of industrial wastewater management: Evaluating challenges and enablers

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ABSTRACT

The study provides a systematic literature review (SLR) encompassing industrial wastewater management research from the past decade, examining enablers, challenges, and prevailing practices. Originating from manufacturing, energy production, and diverse industrial processes, industrial wastewater's handling is critical due to its potential to impact the environment and public health. The research aims to comprehend the current state of industrial wastewater management, pinpoint gaps, and outline future research prospects. The SLR methodology involves scouring the Scopus database, yielding an initial pool of 253 articles. Refinement via search code leaves 101 articles, followed by abstract screening that reduces articles to 79, and finally 66 well-focused articles left for thorough full-text examination. Results underscore the significance of regulatory frameworks, technological innovation, and sustainability considerations as cornerstones for effective wastewater management. However, substantial impediments like; inadequate infrastructure, resource constraints and the necessity for stakeholder collaboration still exist. The study highlights emerging research domains, exemplified by advanced technologies like nanotechnology and bioremediation, alongside the pivotal role of circular economy principles in wastewater management. The SLR offers an exhaustive view of contemporary industrial wastewater management, accentuating the imperative of an all-encompassing approach that integrates regulatory, technological, and sustainability facets. Notably, the research identifies gaps and opportunities for forthcoming exploration, advocating for interdisciplinary research and intensified stakeholder collaboration. The study's insights cater to policymakers, practitioners, and researchers, equipping them to address the challenges and capitalize on prospects in industrial wastewater management effectively.

1. Introduction

1.1. Background of existing knowledge

Water is an essential resource for industrial operations, and its availability, quality and reliability can have significant impacts on business operations and profitability (Li et al., 2019). Industries rely on water for a variety of purposes including; Manufacturing Processes, Power Generation, Agriculture, Mining and Oil and Gas Production etc. (Gingerich et al., 2017). Effective management of water resources is therefore becomes critical for industries, to ensure the sustainability and longevity of their operations (Colella et al., 2021a,b). One of the key challenges in managing water utility in industries is the disposal of wastewater. The discharge of industrial effluents into water bodies can

cause pollution, leading to severe environmental and health consequences (Tong et al., 2022). To address this issue, industries are adopting various wastewater treatment technologies, including physical, chemical and biological treatment methods (Xu et al., 2020). In a nutshell, water utility in industries is crucial for sustainable industrial development, and its efficient management is essential for minimizing the environmental impact of industrial activities. The adoption of water-efficient technologies and effective wastewater treatment methods are crucial for achieving this goal.

There exist various types of industries that heavily depend on water for execution of their respective processes. Like the textile industry requires substantial water for dyeing, printing and finishing processes. A single cotton T-shirt production can consume over 2500 L of water due to these processes (Singh et al., 2019). The paper production involves pulping and bleaching, both of which demand significant water

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Abbreviations		MPs	Micropollutants
SLR	Systematic literature review	C-MFC	Ceramic-microbial fuel cell
WWM	Waste water management	PCPs	Personal care products
WWTPs	Wastewater treatment plants	IoT	Internet of Things
AOPs	Advanced oxidation processes for water treatment	HRAS	High-rate activated sludge
EIDM	Evidence-informed decision-making	CFD	Computational fluid dynamics
PRISMA	Preferred Reporting Items for Systematic Reviews and Meta-Analyses	ICTs	Information and Communication Technologies
CBE	Circular bioeconomy	EAOP	Electrochemical advanced oxidation processes
AMSC	Adaptive Multiple Stakeholder Collaboration	ENMs	Engineered nanomaterials
EDCs	Endocrine disrupting compounds	PFD	Process Flow Diagrams
		LCA	Life Cycle Assessment
		MFC	Microbial fuel cell

quantities. The paper industry is known to be one of the largest consumers of industrial water (Man et al., 2018). Similarly, chemical processes, such as petrochemical refining and pharmaceutical production, require substantial water for cooling, solvent extraction and chemical reactions. Various mining processes, such as ore extraction and mineral processing, necessitate water for dust suppression, material separation and processing (Norgate and Haque, 2010). Coal extraction and preparation activities use substantial water for washing and separating impurities from coal. Hydraulic fracturing (fracking) in the oil and gas industry requires large volumes of water mixed with chemicals to extract hydrocarbons from underground formations. In the same context, thermal power plants (coal, nuclear, and gas) rely on water for cooling purposes (Fthenakis and Kim, 2010). Water is also essential for hydropower and geothermal energy production. Solar panel manufacturing involves water-intensive processes like cleaning, cooling and materials purification. Breweries, wineries, and soft drink manufacturers require substantial water for ingredient preparation, fermentation, and equipment cleaning. Processing agricultural products like fruits, vegetables and grains involves washing, peeling and packaging, which require water. Processing foods like meat, dairy, and fruits often involve washing, rinsing, and cooking, which contribute to significant water consumption (Compton et al., 2018; Sodhi et al., 2020a). These examples highlight the diverse industries reliant on water for their operations. As water scarcity and pollution concerns continue to grow, these sectors must adopt sustainable practices, efficient water use, and innovative technologies to ensure environmental and economic resilience. Addressing these challenges requires collaborative efforts from industries, governments and stakeholders to secure water resources for future generations.

Industrial wastewater is a major environmental concern due to its high pollutant load and potential impacts on water quality and public health (Mekuria et al., 2021). According to United Nations Environment Programme (2017) the problem of industrial wastewater is rising day by day due to several reasons. As industries continue to expand and new ones emerge, the amount of wastewater generated increases, leading to more pollution of water resources. Many industries still lack the proper infrastructure and technology to treat their wastewater adequately, resulting in discharge of untreated or inadequately treated wastewater into water bodies (Trianni et al., 2021). In many regions, there are gaps in regulations or inadequate enforcement of existing regulations, leading to non-compliance by industries (Vrzala et al., 2022). In some cases, monitoring of industrial wastewater is insufficient, leading to limited understanding of the extent of the problem and inadequate action to address it (Rana et al., 2022; Zakaria and Michael, 2017).

Applying deep learning techniques to urban wastewater management presents several challenges that need careful consideration. Two prominent challenges are safeguarding data privacy and addressing multi-agent systems (Saheb and Izadi, 2019; Rippin et al., 2023a). Deep learning relies on large datasets to train models effectively. In the context of urban wastewater management, these datasets may contain

sensitive information, such as operational data, customer usage patterns and infrastructure details. Safeguarding data privacy becomes crucial to ensure that personal and sensitive information is not compromised. However, striking a balance between utilizing valuable data for training and protecting individual privacy can be complex (Tallon, 2013; Sodhi et al., 2020c). Implementing robust data anonymization techniques, encryption methods and access controls are necessary to prevent unauthorized access and data breaches. Ethical considerations surrounding data sharing, informed consent and compliance with data protection regulations besides complicate the integration of deep learning methods.

On the other hand, urban wastewater management involves multiple interacting components, including treatment plants, distribution networks, sensors and actuators. These components form a complex multi-agent system where each agent makes independent decisions to optimize its objectives (Zulkifli et al., 2018). Applying deep learning to such systems requires modeling the interactions, dependencies and feedback loops accurately. Challenges arise in designing deep learning models that can effectively capture the dynamics of multi-agent systems and facilitate collaboration among various agents. The complexity of these systems may lead to challenges like data scarcity for training, scalability issues and interpretability of model outputs (Shen, 2018). Balancing individual agent goals with overall system objectives and designing algorithms for decentralized decision-making are essential for successful application. Elaborating on these challenges emphasizes the intricate nature of integrating deep learning techniques into urban wastewater management. While deep learning holds the promise of revolutionizing this domain, overcoming these challenges is pivotal to ensuring effective, ethical and sustainable solutions.

The rise of industrial wastewater problem has significant implications for the environment and human health. For example, untreated wastewater can lead to contamination of drinking water sources, harming aquatic life, and compromising public health (Ortúzar et al., 2022; Sodhi et al., 2013). Industrial wastewater management is essential for protecting the environment, public health, and business operations (Rahman et al., 2023). The need for effective industrial wastewater management is driven by several factors. In many regions, there are regulations in place that require industries to manage their wastewater in a responsible and sustainable manner. Compliance with these regulations is essential for avoiding fines, legal action and reputational damage (Zan et al., 2022; Rippin et al., 2023b). Proper management of industrial wastewater is essential for protecting water resources, preventing contamination of soils and groundwater and preserving biodiversity (Xi et al., 2022). Untreated industrial wastewater can contain harmful pollutants that can pose risks to public health if they enter the food chain or water supply (Al-Tohamy et al., 2022). Industrial wastewater management is crucial for ensuring the sustainability of business operations, as well as meeting the expectations of stakeholders and customers who demand sustainable practices. Effective industrial wastewater management involves a range of practices, including

Table-1
Perspectives of WWM.

Challenges	Enablers	Specific Issues	Opportunities/Scope
Water Scarcity and Pollution	Advanced Treatment Technologies	Textile dyeing processes release toxic chemicals Mining operations contaminate groundwater Agricultural runoff pollutes water bodies	Membrane filtration for effective color and pollutant removal in textiles Bioremediation techniques for complex organic compounds Closed-loop water recycling in agriculture for water scarcity mitigation and nutrient provision Real-time monitoring systems for timely intervention
Regulatory Compliance and Infrastructure	Digitalization and Monitoring	Complex industrial processes require tailored treatment methods Leather tanning industry releases toxic substances Inadequate wastewater treatment infrastructure	IoT sensors in textile industry for dyeing optimization Closed-loop water recycling in mining for reduced water intake and contamination risks
Deficits	Circular Economy Approaches	High water consumption in mining operations Agricultural wastewater remains untreated	Agricultural wastewater reuse for irrigation and nutrient provision Zero Discharge of Hazardous Chemicals (ZDHC) framework
Collaboration and Knowledge Sharing	Collaboration and Knowledge Sharing Regulatory Frameworks and Incentives	Hazardous chemical use in textiles	Economic incentives for adopting cleaner production processes Regulatory frameworks with fines for non-compliance
Data Privacy and Security	Deep Learning Methods	Sensitive data exposure and breaches Privacy concerns hinder data sharing	Implementation of secure data encryption protocols Secure data sharing platforms with controlled access
Multi-Agent Systems	Integration of AI in Decision-Making Collaborative Platforms and Tools	Lack of standardized communication protocols Lack of coordination among agents	Development of AI-powered multi-agent frameworks AI-driven optimization for efficient resource allocation
Energy Efficiency and Sustainability	Innovative Technologies Policy Support and Funding	High energy consumption in treatment processes Carbon emissions from energy-intensive industries	Integration of renewable energy sources in wastewater treatment Government incentives for energy-efficient technologies
Emerging Contaminants	Research and Development Analytical Methods and Technology	Presence of pharmaceuticals in wastewater Limited analytical methods for detection	Identification of effective treatment methods for emerging contaminants Development of advanced sensors and analytical techniques

wastewater treatment, reuse and recycling as well as, efficient use of water resources and pollution prevention (Patterson, 1985).

According to Atangana et al. (2021) there are several existing wastewater management techniques and practices that are used in industries to treat and manage their wastewater. This involves the use of microorganisms to break down and remove pollutants from wastewater. The most common biological treatment processes include activated sludge, trickling filters, and anaerobic digestion (Verstraete et al., 2022). This includes the use of chemicals such as coagulants, flocculants, and oxidants to remove pollutants from wastewater. Chemical treatment processes include precipitation, oxidation, and adsorption (Zou et al., 2021). It consists of the use of physical processes such as sedimentation, filtration, and membrane separation to remove pollutants from wastewater (Dwivedi et al., 2022). This involves the use of advanced technologies such as reverse osmosis, ultrafiltration, and ion exchange to remove specific pollutants from wastewater (Burge et al., 2020). Table 1 outlines the challenges, enablers, specific issues and corresponding opportunities or scope associated with industrial wastewater management in various sectors.

The above table aims to offer a comprehensive overview of the complexities and potential solutions in addressing these challenges across different industries and sectors.

In addition to these techniques, there are also several practices that industries can adopt to manage their wastewater more effectively. Water conservation reduces the amount of water used in industrial processes through measures such as; process optimization and reuse (Cao and Yang, 2020). Pollution prevention implements the measures to prevent or reduce the generation of pollutants in the first place, such as process modification and use of cleaner technologies (Chai et al., 2021). Similarly sustainable wastewater management adopts a holistic approach to wastewater management that considers the entire lifecycle of wastewater, from generation to disposal.

1.2. Research Gaps

There is growing awareness among industries about the importance of wastewater management and the need to adopt sustainable practices (Frkova et al., 2020). Many industries are recognizing the potential benefits of effective wastewater management, including reduced environmental impacts, improved public health, and enhanced business sustainability (Harahap et al., 2021). However, there are still significant challenges that need to be addressed to ensure that industries are fully aware of the importance of wastewater management and are taking effective measures to manage their wastewater (Vystavna et al., 2018). These challenges include the lack of regulatory enforcement in some regions, limited resources for wastewater management and a lack of understanding of the potential impacts of poor wastewater management practices (Lasode et al., 2021). Moreover there are several reasons why existing wastewater management techniques may fail or be ineffective in achieving desired outcomes. Many wastewater treatment facilities use outdated or insufficient treatment technologies that are unable to effectively remove pollutants and contaminants from the wastewater. As a result, untreated or partially treated wastewater is released into the environment, leading to environmental degradation and health risks. Even the most effective wastewater treatment technologies can fail if they are not properly maintained and operated (Nunes, 2021). Inadequate maintenance can result in equipment breakdowns, clogging of pipelines and other issues that can lead to reduced treatment efficiency and increased operational costs (Gulfam-E-Jannat et al., 2023). Many wastewater treatment facilities, especially in developing countries, lack the financial resources to invest in modern and effective treatment technologies and infrastructure (Rodrigues de Assis et al., 2020). This limits their ability to effectively treat wastewater and prevent pollution. In some cases, the failure of wastewater management techniques may be attributed to a lack of public awareness and engagement. Without public support and participation, it can be challenging to implement and sustain effective wastewater management practices (Burge et al., 2020).

Sometimes, existing regulations may not be sufficiently robust or enforced to ensure effective wastewater management (Khan et al., 2020). This can result in non-compliance by industrial operators and other polluters, leading to negative environmental and health impacts.

Rice et al. (2012) stated that underutilization and operational inefficiencies are major causes of failure in wastewater treatment plants (WWTPs). WWTPs that use outdated or inadequate treatment technologies may struggle to effectively treat wastewater (Karaolia, 2019). This can result in reduced treatment efficiency, increased operating costs, and higher rates of equipment breakdowns (Stamatelatos and Tsagarakis, 2015). WWTPs require trained and knowledgeable staff to operate effectively. Inadequate staff training or insufficient resources can lead to reduced treatment efficiency, increased operating costs, and higher rates of equipment failures (Skoulidakis and Ganoulis, 2012). WWTPs require ongoing investments in infrastructure, equipment and technology to remain effective (Tili et al., 2020). Limited financial resources can lead to underinvestment in critical areas, reducing treatment efficiency and increasing the risk of equipment failures. WWTPs require ongoing monitoring and evaluation to ensure that treatment processes are working effectively and that treatment goals are being met (Magro et al., 2020). Inadequate monitoring and evaluation can lead to operational inefficiencies and reduced treatment efficiency (Petcharat et al., 2020).

Next to focus logically, an 'Umbrella Review' has been performed to identify and critically evaluate all of the available Review Papers (may be narrative, systematic or meta-analyses reviews) on the topic of "Waste Water Management" published in last decade, for synthesizing their respective findings and to draw rational conclusions about the overall state of the evidence. This approach allows for a more comprehensive and nuanced understanding of the available evidence, as well as the strengths and limitations of individual studies (Moggio et al., 2022; Ioannidis, 2016). The primary aim of conducting the "Umbrella Review" is to provide a comprehensive and synthesized overview of existing review papers in the field of industrial wastewater management. By systematically analyzing and synthesizing these review papers, our intention is to identify common themes, emerging trends, gaps in knowledge and potential areas for future research. The data captured during umbrella review has been analyzed to bring advance implications to design an effective SLR. An Umbrella Review of existing 28 review papers (found on Scopus database in last decade) have been appropriately quoted in Table 2 ahead.

The table presents a summary of review-papers published from 2013 to 2022 on different topics related to water and wastewater management. The review-papers have been analyzed based on critical aspects like; their authors, year of publication, research area, the number of papers reviewed, the method/procedure used, limitations observed and cited by number of times, respectively. The reviews cover various research areas, including the digitalization of wastewater management, advanced oxidation processes, priority substances in urban wastewater, emerging nanomaterials for water purification, degradation of organic pollutants, selective oxidation of contaminants, solar energy for wastewater treatment, implementing advanced micro-pollutants removal technologies and chemico-nanotreatment methods for removing persistent organic pollutants & xenobiotics in wastewater treatment plants (see Table 2). Other studies assess technologies for treating water with ionic liquids, the presence of nano/micro-plastics in water and wastewater treatment processes, incorporating green energy and energy-efficient technologies for municipal wastewater treatment plants, and advancements in hydro-metallurgical technologies for recovering critical raw materials and precious metals from low-concentration streams.

Different types of reviews have been conducted on WWM, such as narrative review, critical review, rapid review and scoping review. In the field of academic literature, these types of reviews serve specific purposes in contributing to a comprehensive understanding of a topic. A narrative review provides a broad overview of existing literature, often summarizing key findings without a systematic approach. It aims to

present a general understanding of the topic and highlight trends or important studies. A critical review involves a detailed assessment and analysis of selected studies. It not only summarizes findings but also critically evaluates their methodology, limitations, and implications. This type of review offers a nuanced perspective on the quality of evidence. A rapid review focuses on summarizing a large amount of literature in a shorter timeframe. It aims to provide timely insights by streamlining the review process while still adhering to systematic principles. A scoping review maps the available literature to identify the breadth and depth of a topic. It helps to clarify research questions and identify gaps in knowledge before conducting a more comprehensive systematic review. Each type of review contributes uniquely to the understanding of wastewater management. A narrative review can offer an introductory overview, a critical review dives deep into specific aspects, a rapid review provides quick insights, and a scoping review helps to define the scope of research. By employing these different review types, our analysis encompasses a wide range of perspectives, ensuring a holistic understanding of industrial wastewater management and its challenges and enablers.

Some authors did not follow any specific scientific method, which led to limitations in their scope and conclusions. For example, some papers focused on a specific region or pollutant and did not consider other relevant aspects. One research paper analyzed the presence and elimination of medications in African water sources, primarily in urban regions. Another study assessed hydroponic technology as a method for decentralized domestic wastewater treatment and vegetable cultivation in urban farming, though it exaggerated the technology's advantages and disregarded practical limitations. A comprehensive evaluation of municipal wastewater treatment in China examined the evolution and prospects of the field, but lacked a proper synthesis and conclusion.

In last decade the number of papers reviewed range from 28 to 315. The citation counts of the papers varied from 0 to 544. Some limitations observed in the papers included incomplete coverage of the topic, lack of comprehensiveness and the use of informal sources. In conclusion, the research papers highlighted the challenges and opportunities in the field of water and wastewater management and provided insights into various technologies and strategies for effective water and wastewater management (see Table 2 for details).

Some studies have limitations, such as a lack of transparency and replicability, bias and dependence on author perception, a lack of quantitative comparison, less explicit methods, and high risk of bias and incompleteness. Overall, the studies highlighted the need for more research and development in wastewater treatment technologies, as well as emphasize more on the importance of considering real-world constraints and limitations while implementation.

1.3. Implications to design an effective SLR

Above review data-table is further analyzed through Minitab Statistical Software. Firstly Chi-Sq Test has been applied to find the 'relative impact' of various types of Review Papers Published in the field of Waste Water Management. As the p-value is coming out to be less than 0.05 barrier, Hence it can be concluded with 95% confidence that there exists statistically significant difference in the Number of Review Papers Published w.r.t. the 'Type of Review' not only in last decade but this trend is beyond that. (As far as WWM papers are concerned). Around 12 Critical Reviews have been published on WWM, whereas 8 Narrative Reviews and 5 Rapid Reviews have been written. Approximately 3 Rapid and Scoping Review Papers has been published on WWM. Only 1 Systematic Literature Review and 1 Umbrella Review Paper has been published on Waste Water Management since 2012. Critical Review Papers have found maximum impact in WWM Literature. Though having only 1 Systematic Review Paper in last decade, But still it has found to have more contribution to the Ch-Sq value as compared to Narrative, Rapid and Scoping Reviews (refer Fig. 1).

It has reviewed less (88) Papers and still have greater Citations (385).

Table-2
Umbrella review.

Sr. No.	Authors with Year of Publication	Research Area	Papers Reviewed	Method/ Procedure	Limitations Observed	Cited by
1	Fu et al. (2022).	Digitalization of WWM in cities	144	Narrative Review	The study was limited to waste water management in urban cities only. Important waste water issues like; strategic planning, maintenance and its continual development were totally ignored.	0
2	Priyadarshini et al. (2022).	Advanced oxidation processes for water treatment (AOPs)	240	Critical Review	Lack of comprehensiveness was observed as in case of AOPs very less scale up trials had been considered. Work on Technoeconomic assessments, optimization of reactor design solar light innovations and upcoming catalytic technologies were partially touched.	9
3	Birtek et al. (2022).	Occurrence and risk estimation of priority substances in urban wastewaters	127	Critical Review	It doesn't talk about the treatment of priority or banned substances in waste water and restricted its study only for Istanbul.	0
4	Yu et al. (2022).	MXenes as nanomaterials for environmental remediation	133	Critical Review	Not followed any standard review procedure like PRISMA, ROSE or CEE Guidelines. Moreover nothing was discussed to handle secondary pollution caused by Mxenes due to further release of adsorbed contaminants.	151
5	Subhiksha et al. (2022).	Bismuth-based photocatalysts for degradation of organic pollutants	182	Critical Review	This review approach is considered conventional and assumes that all studies are equally valid. The review found a limited amount of literature regarding the mobility of charge carriers, insufficient bandgap, inadequate ability to separate charges, and limited capacity for absorbing light in BBPs.	17
6	Yang et al. (2021).	Selective oxidation of contaminants in aqueous systems	102	Critical Review	Realized citations from number of informal sources like web pages or sites, whose credibility should be mentioned. Depicted quite a few applications of selective oxidation in real scenario.	32
7	Pandey et al. (2021).	Utilization of solar energy for wastewater treatment	118	Rapid Review	Covered techniques utilized since last decade only. Not many studies were cited about the catalyst which is essential part of solar powered degradation of pollutants. More elaborative discussion was needed to improve the performance, stability and capabilities of solar based waste water treatment plants.	29
8	Karthigadevi et al. (2021).	Chemico-nanotreatment for removal of persistent organic pollutants and xenobiotics	233	Critical Review	Only Xenobiotic contamination was focused since last decade. Not followed any specific scientific method, which inculcate unwanted characteristic of non-replication in the review. Scope was also less emphasized in conclusion section.	25
9	Kataki et al. (2021).	Concerns and strategies for wastewater treatment during COVID-19 pandemic	175	Scoping Review	Limited literature on effective treatment of viral contaminated wastewater led to a non-rigorous review. The study suggests using multiple treatment barriers for complete removal of viruses from wastewater and taking extra precautions.	52
10	Kosek et al. (2020).	Advanced micropollutants removal technologies in WWTPs	61	Narrative Review	Focused on micropollutant removal in Europe, suggesting upgrades to certain urban WWTPs, but did not address cost-effective elimination of a wide range of micropollutants.	37
11	Ye et al. (2020).	Pollution control using artificial intelligence	249	Scoping Review	The review focused on recent adoption of AI was based on a few case studies. The AI proposed model relied on empirical experience and only validated with historical data.	84
12	Wang et al. (2020).	Fugacity modelling of micropollutant fate in aqueous systems	94	Critical Review	Deals with specific technology and the study were observed more descriptive in nature, instead of integrating or synthesizing revolutionary outcomes.	10
13	Madikizela et al. (2020).	Analysis, occurrence, and removal of pharmaceuticals in African water resources	95	Narrative Review	Search Strategies were not mentioned in details. It was more or less a qualitative review and focused in urban African areas mostly.	66
14	Magwaza et al. (2020).	Hydroponic technology for decentralized wastewater treatment and vegetable production	118	Scoping Review	Seems to be overstating the strengths of Hydroponics and understating the real world constraints. Proposed approach was appropriate for decentralized wastewater treatment only.	61
15	Qu et al. (2019).	Municipal wastewater treatment in China	28	Critical Review	Qualitative aspect of China's waste water management had been delineated without any proper synthesis and conclusions.	174
16	Isosaari et al. (2019).	Ionic liquid-based water treatment technologies for organic pollutants	111	Narrative Review	The study seemed to be re-hashing what's already been said. Lack of integration and synthesis had been realized.	79
17	Enfrin et al. (2019).	Nano/microplastics in water and wastewater treatment processes	191	Umbrella Review	Study conducted with sufficient number of papers but still lacking in systematic execution of a reviewing process. Did not mentioned assumptions or limitations of present study.	225
18	Guo et al. (2019).	Integration of green energy and energy-efficient technologies for WWTPs	189	Rapid Review	Emphasized more on energy savings by bringing optimality in design and operational aspects of plants rather than relying on some sustainable energy alternatives.	33
19	Perez et al. (2019).	Progress in hydrometallurgical technologies for resource recovery	108	Narrative Review	No quantitative comparison of technologies was provided, and important factors such as operational costs, environmental impact and low efficiency in diluting metal concentrations were not emphasized.	54
20	Sancho et al. (2019).	Carbon redirection in wastewater treatment plants	75	Critical Review	Study covered only last 10 year literature and lacked repeatability as it was not followed any systematic standardized approach of reviewing.	57
21	McCance et al. (2018).	Contaminants of Emerging Concern as groundwater tracers in urban areas	142	Rapid Review	A lack of transparency/replicability might be the major concern. Referred a few latest or upcoming papers.	66
22	Morris et al. (2017).	Municipal wastewater effluent licensing and best practices	105	Critical Review	Discussed only the Licensing importance to protect waste water, though emerging contaminants were still out of the regulations.	20

(continued on next page)

Table-2 (continued)

Sr. No.	Authors with Year of Publication	Research Area	Papers Reviewed	Method/ Procedure	Limitations Observed	Cited by
23	Chen et al. (2017).	Regulation strategies of microbial community dynamics in wastewater treatment systems	99	Narrative Review	Found to be biased and partially depended upon the author's perceptions.	92
24	Zhang et al. (2016).	status of WWM in China	52	Narrative Review	Delineated review from Chinese publications with high risk of bias and incompleteness	332
25	Gerbersdorf et al. (2015).	Anthropogenic trace compounds in aquatic habitats and elimination strategies	155	Critical Review	Less explicit methods had been used and without any Meta-Analysis for quantitative findings.	66
26	Chaplin (2014).	Electrochemical advanced oxidation processes for water treatment	315	Critical Review	Seemed to be non-comprehensive rapid study which ignored many real world constraints. Emphasized more on electro chemical part of oxidations.	544
27	Neale et al. (2013).	Detection, fate, and effects of engineered nanomaterials in wastewater treatment plants	92	Narrative Review	Used non-explicit methods with quick conclusions having high risk of bias and possible errors.	34
28	Corominas et al. (2013).	Life cycle assessment of wastewater treatment processes.	88	Systematic Review	Lack of evidence-informed decision-making (EIDM)	385

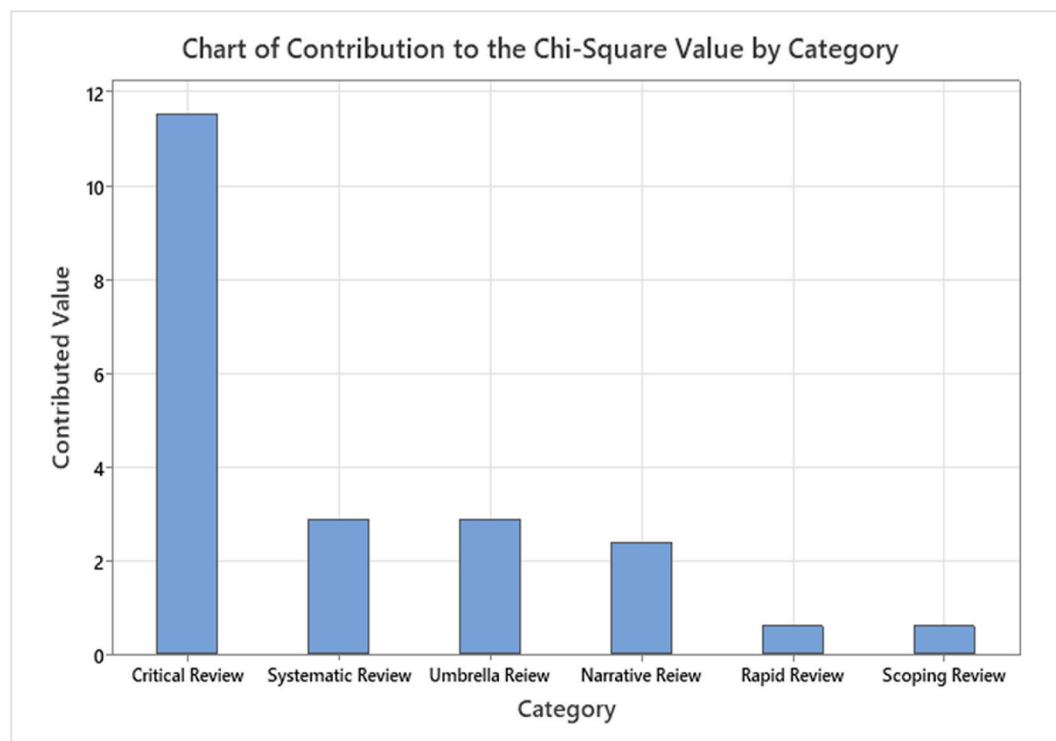


Fig.1. Contribution to WWM literature.

This depicted the potential of Systematic Review Paper and put light on its acceptability, clarity and effectiveness. It is obvious from above figure that though systematic literature reviews (SLRs) are generated from less number of references, but still their comparative citations are impressively high. This fact uncovered the hidden potential of SLR and motivates further to perform a comprehensive systematic review without ignoring its descriptive part. Overall, the methodological rigor, comprehensive coverage and trustworthiness of SLRs contribute to their high citation count compared to other types of review papers. Researchers value SLRs as authoritative references that offer reliable insights, help bridge knowledge gaps, and stimulate future research in the field. SLRs play a critical role in evidence-based practice by providing reliable summaries of existing research. Researchers and practitioners from diverse fields reference SLRs to inform their decisions and practices. While Systematic Literature Reviews (SLRs) offer valuable insights, potential limitations should also be acknowledged. These include publication and language biases, database selection, varying study

quality, subjective inclusion criteria and timeframes affecting relevance, potential errors in data extraction and synthesis, overlooking grey literature, expertise bias and capturing only up-to-date trends. Recognizing these limitations ensures a balanced interpretation of findings, stresses the need for cautious conclusions and highlights areas for future research. This transparency enhances the credibility and reliability of SLRs while guiding further exploration in the field.

Best Review Paper is the article which everyone wants to read and cite in their own work. Citations play a crucial role in assessing the significance and influence of a review paper. High citations indicate that the paper has been widely recognized and referenced by other researchers, suggesting that it has contributed substantially to the field's knowledge base. Such recognition signifies that the review paper has influenced subsequent research, discussions and developments. In order to design an appropriate Systematic Review, necessary recommendations can be taken after regressing existing independent factors (Xs) with corresponding Y (i.e. Citations; in which we are interested). Presently,

the Year of Publication, Number of Papers Reviewed and Method (Type) of Review have been considered as Independent Factors (Xs), which can affect the Dependent Factor (Y) i.e. Citations of the Publication. An effort has been made to regress relationship between Citations (Y) and with above Xs by using last ten year WW Review data. Data is fed into Minitab for appropriate Model fitting and found following results. The p-value less than 0.05 depicted the statistical significance of all the considered Xs to raise the overall citations of a WWM Review Papers (see Fig. 2). The R-sq (79.8%) proved the efficacy of quadratic model on present review dataset. The 3 plots of 'Cited by' Versus 'X variables' are having non-grey background and their scatter plots reflected sufficient importance of factors like; Year of Publication, Number of Papers Reviewed and Method (or Type of Review) performed. These finding implies that Year of Publication contributes to citations, underlines the importance of staying up-to-date with the latest research trends. Incorporating a higher number of papers reviewed enhances the comprehensiveness and depth of the review, potentially attracting more citations as a valuable resource. Selecting the appropriate method of review (such as systematic or critical approaches) can also impact citations by ensuring rigorous methodologies and in-depth analyses that contribute to the review's credibility and authority. Therefore, these findings provide insights for researchers aiming to design impactful SLRs, guiding them to strategically consider these factors to maximize the potential for higher citations and broader influence in the field of wastewater management.

Model Building sequence illustrated the relative importance of all the independent factors (Xs) along with their interactions (see the corresponding p-value for significance in Fig. 2). Like square of Papers Reviewed are more important ($p = 0.001$) than Paper Reviewed as a factor. Similarly Year of Publication and Type of Review are not having

much significance, but their interactions with Number of Papers Reviewed are statistically important. Relative Importance of under studied factors are drawn with help of linear bar graphs, showing Year of Publication has least significance, while Number of Papers reviewed has maximum influence on the Citations (Y). The bar-graph reveals that Year of Publication has the least significance, suggesting that the age of a paper does not necessarily dictate its impact or influence. On the other hand, the Number of Papers Reviewed emerges as the factor with the highest influence on Citations (Y), indicating that the thoroughness and comprehensiveness of the review process play a pivotal role in attracting citations. This highlights the importance of conducting comprehensive literature searches and including a substantial number of relevant papers in systematic literature reviews. Researchers and authors should prioritize the depth and breadth of their reviews to enhance the likelihood of higher citations. Moreover, these implications highlight the dynamic nature of research impact, where a comprehensive and up-to-date approach can outweigh the effect of publication year in contributing to a review's overall citations. The above facts will be quite helpful to design a good systematic review-paper on WWM, which will have maximum number of citations and popularity.

The citations received by WWM Review-Papers have been delineated with the help of a line diagram (see Fig. 3). The observed behavior of citations in systematic literature reviews (SLRs) reveals a distinct trend. Initially, as the number of reviewed papers rises from 25 to 170, citations decrease. Beyond this range, with over 200 reviewed papers, citations sharply increase, reaching 150 or more. This suggests that reviews with a higher number of papers reviewed tend to attract more citations. This phenomenon may be attributed to improved comprehensiveness, coverage and authority of the review, making it a reliable reference for researchers. A comprehensive review with over 200 papers

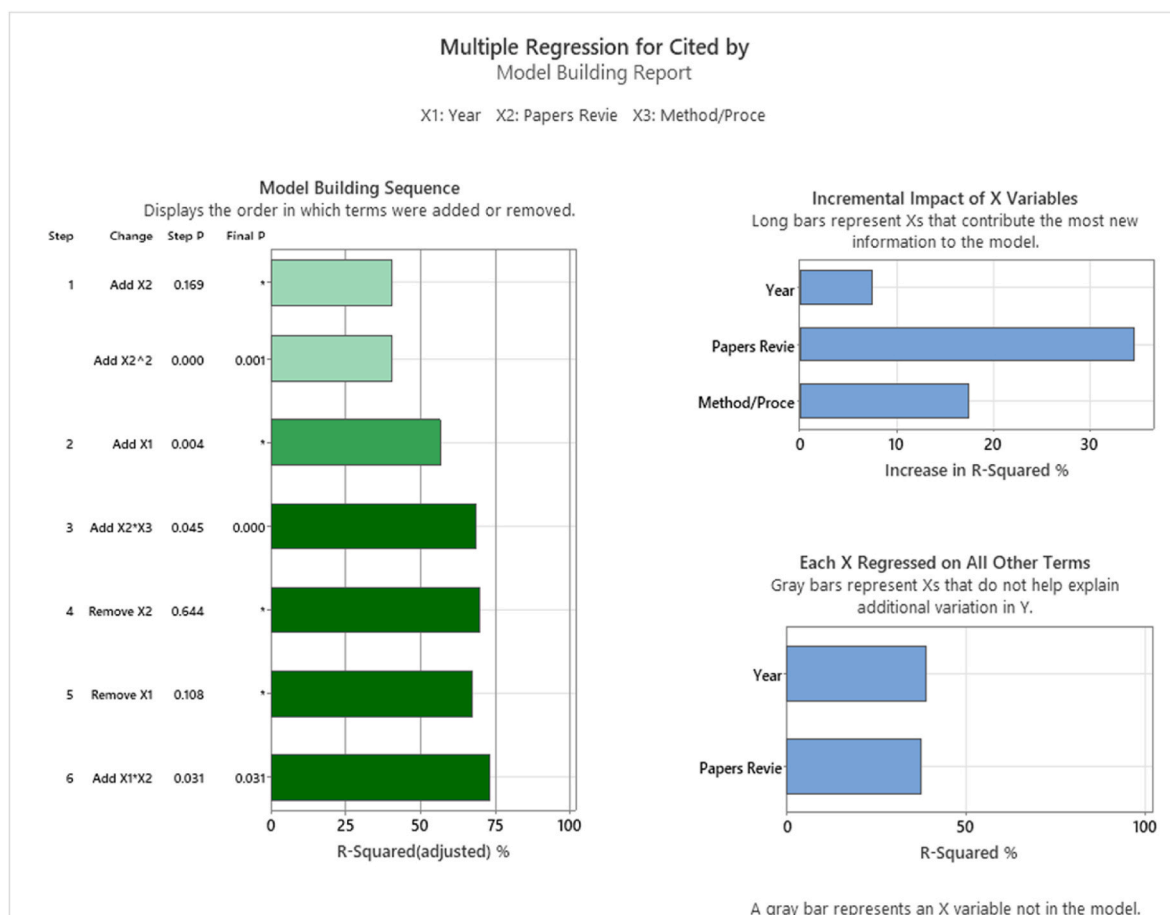


Fig. 2. Regression model.

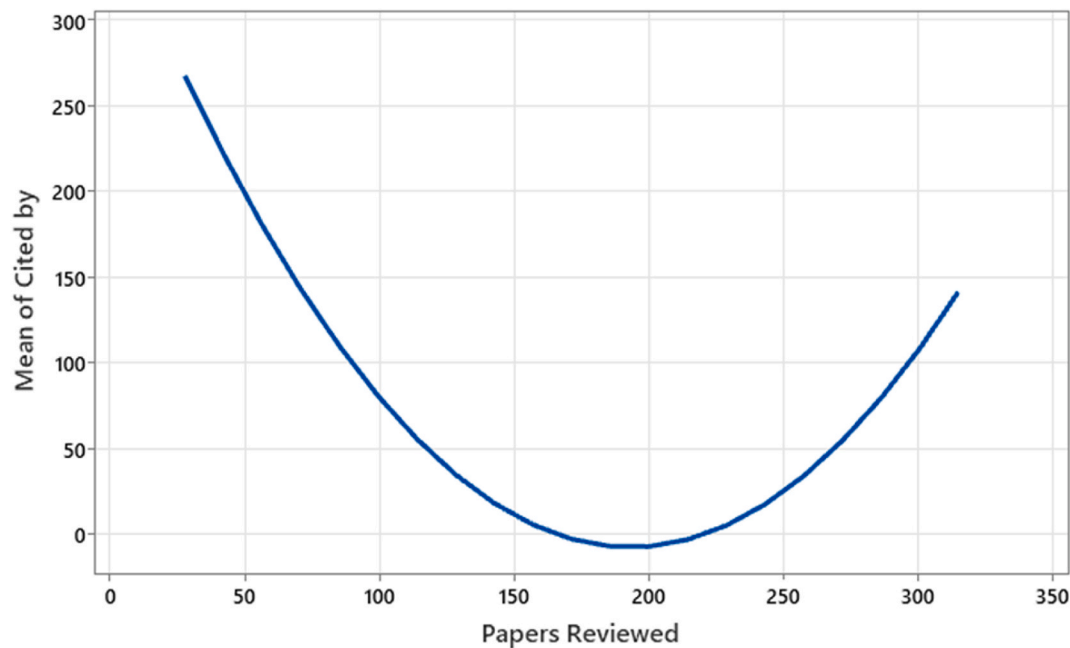


Fig. 3. Main effect plot for citations received Vs number of papers reviewed.

seems to be perceived as more authoritative, driving higher citations and impact within the research community.

Minimizing bias is imperative in conducting a systematic literature review (SLR). Ensuring that the review process is free from bias is essential to maintain the credibility and validity of the findings. Bias can arise from various sources, such as selection bias when choosing studies, publication bias in favor of positive results, and interpretation bias in analyzing the data. To mitigate bias, researchers must establish clear inclusion and exclusion criteria, use comprehensive search strategies, and transparently document the review process. Additionally, employing multiple reviewers and resolving disagreements through consensus can help reduce individual bias. By taking these precautions, the SLR results are more likely to accurately reflect the true state of the literature, enhancing the overall quality of the review.

Increasing the validity and reliability of a review is essential to enhance the credibility of its findings. Validity ensures that the review accurately represents the true state of the literature, while reliability ensures that the review can be replicated with consistent results. Validity is crucial because it assures readers that the conclusions drawn from the review are trustworthy and relevant to the research question. Reliability, on the other hand, ensures that the review process is transparent and unbiased, enabling other researchers to replicate the study and confirm its results. By prioritizing validity and reliability, the review gains greater trustworthiness and becomes a valuable resource for decision-making, policy development, and further research in the field. Next, the utilization of systematic and rational methods to synthesize the results of multiple studies offers several advantages that contribute to a more robust and accurate understanding of the evidence. By applying a systematic approach, researchers ensure that all relevant studies are included, minimizing the risk of bias and selective reporting. This comprehensive inclusion enhances the reliability of the conclusions drawn from the synthesis. Additionally, the rational method allows for the aggregation of findings across studies, providing a clearer and more nuanced overview of the patterns, trends and inconsistencies present in the literature. This approach facilitates the identification of common themes and differences, enabling researchers to make more informed assessments and draw meaningful insights. Ultimately, employing rational methods enhances the credibility and reliability of the synthesized results, making them more valuable for informing decisions,

shaping policies and guiding future research directions.

SLRs are esteemed for providing higher-level evidence due to their rigorous and systematic methodology, encompassing thorough literature searches, quality assessments, and standardized data extraction. This meticulous approach reduces bias and ensures comprehensive synthesis, enhancing the credibility of deductions. Policymakers and practitioners value SLRs as reliable sources that inform decision-making and policy formulation across various domains, including wastewater management. So, a strong need of conducting a rigorous and transparent SLR for identifying, selecting and analyzing relevant studies has been felt. This helps to minimize bias and increase the validity and reliability of the review. It must include the use of rational methods to synthesize the results of multiple studies, providing a more precise estimate of the effect size of an intervention or exposure. A SLR illustrating higher level of evidence than any other type of reviews is observed to be the need of the hour.

1.4. Research questions (problem statements)

After critically examining the existing knowledge about the WWM and its corresponding limitations, it has been decided to explore two major aspects in this manuscript quoted as below.

- **Review the various potential Challenges faced by Industrial WWM Systems.**
- **Assess the essential Enablers to deploy WWM efficiently.**

Understanding the various challenges associated with waste water management is crucial for developing effective solutions to cope with the rising need for waste water treatment (Efeovbokhan et al., 2019). Waste water treatment involves a complex set of processes that can be influenced by a variety of factors, such as the type and quantity of pollutants, the characteristics of the waste water, and the capacity and efficiency of treatment plants. Understanding these factors is essential for developing appropriate treatment strategies (Yang et al., 2017). Waste water management can be expensive, requiring significant financial and human resources (Boguniewicz-Zablocka et al., 2018). Understanding the costs and benefits of different treatment options can help to prioritize investments and maximize the efficiency of existing

infrastructure (Barton et al., 2018). Waste water treatment can have both positive and negative environmental impacts (Pau et al., 2013). Understanding these impacts is important for ensuring that waste water management is sustainable and minimizes harm to the environment (Guldhe et al., 2017). Poor waste water management can have serious public health and safety implications, such as the spread of waterborne diseases and contamination of drinking water sources. Understanding these risks is important for developing effective management strategies to protect public health (Reichwaldt et al., 2017).

Emphasizing on essential enablers is equally important for deploying Waste Water Management (WWM) efficiently in industries, because it can help to overcome some of the common challenges associated with WWM (Amador et al., 2018). Compliance with regulatory requirements is essential for ensuring that WWM practices are effective and meet minimum environmental and public health standards. Essential enablers, such as regulatory frameworks and policies, can provide guidance and support to industries to meet these requirements (Hjelmsø et al., 2017). Advances in technology can enable more efficient and effective WWM practices, such as the use of advanced treatment technologies and real-time monitoring systems. Essential enablers, such as research and development, can drive innovation and encourage the adoption of new technologies (Schäfer et al., 2017). Collaborations between the public and private sectors can help to overcome resource constraints and improve the efficiency of WWM practices. Essential enablers, such as public-private partnerships, can facilitate knowledge sharing, funding, and other resources to support the development and implementation of WWM strategies. Effective WWM practices require trained personnel with the necessary technical knowledge and skills to manage and operate treatment systems (Ozor and Mbohwa, 2018). Essential enablers, such as education and training programs, can help to build capacity and develop a skilled workforce to support WWM practices (Keeley et al., 2013). In summary, emphasizing essential enablers is critical for deploying WWM efficiently in industries, as these can provide the necessary support and resources to overcome challenges and ensure the effectiveness of WWM practices.

1.5. Structure of SLR

This paper includes background and context of industrial wastewater generation, significance of industrial wastewater management and formulates the research objectives and scope after analyzing the existing gaps. The literature review has been performed to define and categorize the industrial wastewater, environmental and public health impacts of unmanaged industrial wastewater and previous research on industrial wastewater management. The methodology is based on well-structured PRISMA approach, which includes inclusion and exclusion criteria for article selection, Search strategy and database (Scopus) utilization, refining article selection through search code, abstract screening and full article exploration. Synthesis of findings regarding enablers, challenges and emerging areas would be made without ignoring the importance of a holistic approach integrating regulatory, technological and sustainability aspects. This paper will act as a whistle blower, which will identify gaps and opportunities for future research and will provide necessary recommendations, as far as industrial WWM is concerned. Required recap of key findings and contributions will be quoted in conclusion section. The citations for all the articles reviewed will be mentioned in the reference list at end.

2. Method

Without ignoring the above implications to design appropriate SLR and research questions, it was decided to write a unique SLR with a blend of descriptive along with qualitative analytics. This is a unique approach with some modifications to well tested Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) approach. PRISMA is a widely accepted framework for conducting and reporting

systematic literature reviews. The PRISMA approach provides a structured and transparent process for identifying, screening, and selecting relevant studies, as well as assessing the quality of the evidence and synthesizing the findings (Moher et al., 2009). The modified PRISMA approach applied in present study is delineated suitably in Fig. 4 and contains the following sequential steps:

Step-1: Research Questions: The first step is to outline the research question and identify the scope of the literature review.

Step-2: Search Strategy: An inclusive search strategy is developed to identify relevant studies from best-fit databases (like SCOPUS, Web of Sciences etc.). Refine the results by applying significant filters on the database.

Step-3: Descriptive Analysis: Descriptive statistics are used to summarize the characteristics of the selected studies, including study design, sample size, and outcome measures.

Step-4: Literature Classification: Literature classification through text mining is a method of analyzing a large corpus of text data, such as academic papers or books, to identify patterns and topics that emerge from the language used in the texts. A machine learning based software namely 'VOSviewer' is used in present case.

Step-5: Record Screening: Inclusion and exclusion criteria are developed to select studies that meet the research question and objectives after examining corresponding abstracts of the Papers.

Step-6: Quality (Eligibility) Assessment: The quality of the selected studies is assessed using predetermined criteria and after thorough study of whole manuscript.

Step-7: Qualitative Analysis/Synthesis: Relevant text is extracted from the selected studies and organized into a tabular form for qualitative analysis and inferences.

The PRISMA approach is widely used in health and medical research to ensure the quality, reproducibility and transparency of systematic reviews (Page et al., 2021). By combining multiple methods of analysis, the proposed approach can provide a comprehensive overview of the existing research on Waste Water Management and identify key issues or areas for future research.

3. Findings

3.1. Step-1

The research questions have been raised in the previous section. Publications from Scopus data base have been shortlisted for carrying out the further study. During the systematic literature review (SLR) process, inclusion and exclusion criteria are employed to guide the screening of studies and ensure that only relevant and high-quality research is considered. Inclusion criteria define the characteristics that studies must possess to be included in the review, such as specific research areas, study designs, publication dates, and relevant outcomes. Exclusion criteria, on the other hand, outlines the reasons for excluding studies that do not meet the predefined criteria. These criteria help researchers focus on studies that align closely with the research question, while eliminating studies that may not contribute substantially or may introduce bias (Rippin et al., 2022). This systematic approach enhances the rigor and reliability of the review by ensuring that only appropriate and reliable evidence is included in the analysis. In the present case, last ten year manuscripts related to waste water management have been explored. This SLR considered only papers published in English language. Because of having full institutional access to full-text articles, no obstacle was faced to access subscription-based journals. But exclusion of unpublished or grey literature (or papers in press) could result in missing out on valuable insights.

However, several studies have compared the performance of Scopus and Web of Science databases in terms of coverage, search functionality and citation metrics (Ge et al., 2018). It has been found that Scopus has certain advantages over Web of Science for conducting systematic literature reviews. For example; Scopus has a larger and more diverse

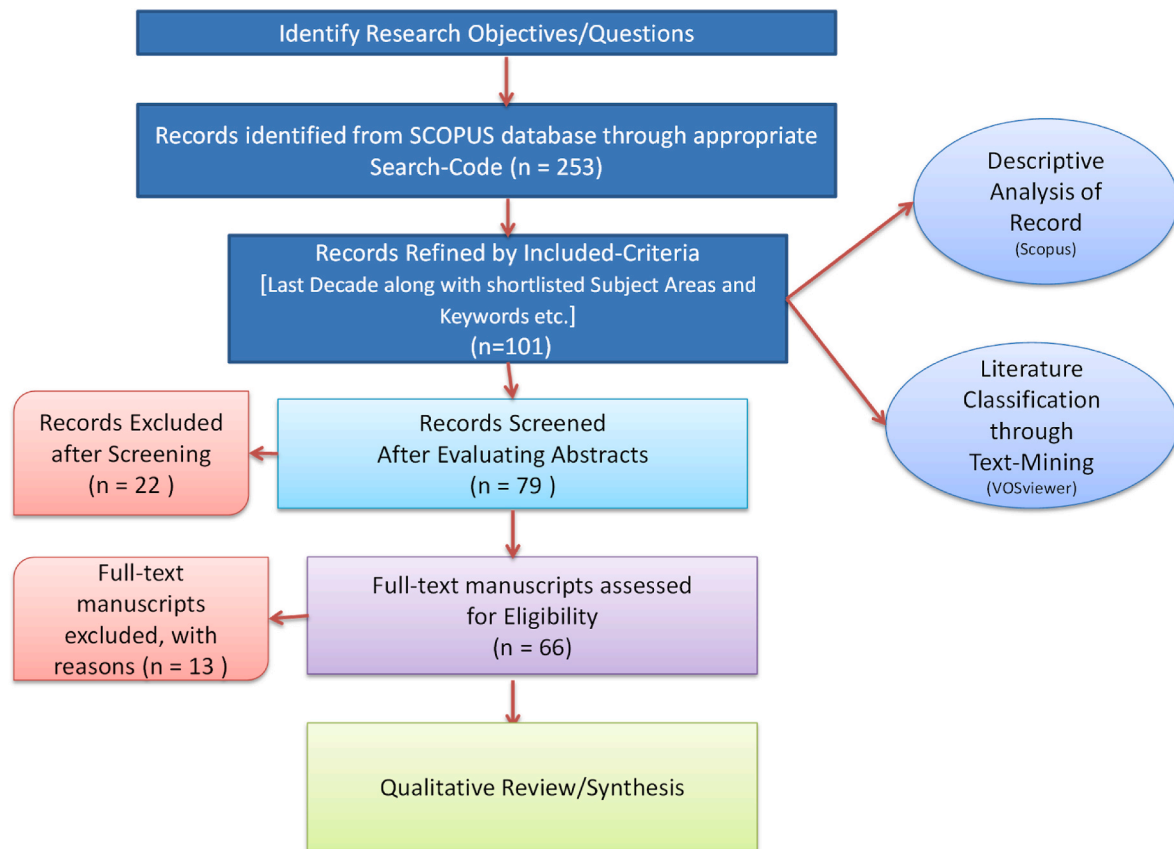


Fig. 4. Proposed methodology.

collection of indexed journals and conference proceedings than Web of Science, covering more disciplines and regions (Falagas et al., 2008; Crow and Murray, 2020). Scopus provides more comprehensive search functionality, including advanced search options, citation tracking, and author and affiliation searching, which can enhance the precision and recall of the search results (Jacso, 2005; Harzing, 2013; Silombela et al., 2018). Scopus includes more non-English language sources, particularly in the social sciences and humanities, which can increase the international and interdisciplinary scope of the review (Mongeon and Paul-Hus, 2016). Scopus has more reliable and transparent citation metrics, such as the h-index and the CiteScore, which can be used to evaluate the impact and visibility of the included studies (Falagas et al., 2008; Mongeon and Paul-Hus, 2016).

The search strategy and selection process are meticulously designed to align with the research objectives of the systematic literature review. By systematically identifying, screening and including relevant studies, these steps ensure that the final selection is comprehensive and representative of the research landscape. This approach contributes to the overall goals of the review by providing a robust foundation of evidence for analysis, enabling the synthesis of findings, and offering meaningful insights to address the research questions and objectives effectively. To ensure the reliability and validity of the selected studies, rigorous quality assessment criteria were applied during the screening and inclusion process. This approach helped to enhance the credibility of the included studies and maintain the overall integrity of the systematic literature review (Sodhi et al., 2020b; Nudurupati et al., 2022). Additionally, proper discussions will be conducted at the end, to assess the potential impact of publication bias on the overall findings, their verification and validation by comparing with the already performed studies will further enhance the robustness of the review's outcomes. The specific reasons for exclusion of certain studies would primarily be based on their lack of relevance to the research objectives, limited scope or

inadequate methodology. Studies that would focus solely on unrelated topics or employed non-rigorous methodologies will be excluded to ensure the integrity and quality of the selected studies.

3.2. Step-2

An appropriate search code has been drafted and added as an input to harness suitable publications, without ignoring the gaps of existing literature and review objectives (already formulated) for the present study (refer Fig. 5 for details).

Around 253 publications on waste water management have been skimmed from Scopus database by using above search code (designed as per database terminology). In order to create more focused list of publications, suitable refine filters have been applied (see Annexure 1 for more insights).

By taking care of implications for designing proper SLR (quoted in section 1.3 above), suitable refine filters like; Subject Area Covered, Document Type, Publication Stage, Source Title, Keywords, Funding Sponsors and Country, respectively have been used, which additionally funneled down 101 potential publications, complying the devised research plans. The refine filters were strategically chosen to target specific aspects of industrial wastewater management, such as treatment technologies, environmental impact assessment, and regulatory frameworks (refer annexure 1). These filters aimed to narrow down the search results to studies that align closely with the research objectives. By applying refine filters that are tailored to the key themes of the review, the study sought to ensure that the selected publications are highly pertinent to the scope of industrial wastewater management. The search code utilized to retrieve publications from the Scopus database was formulated by strategically combining above keywords and terms that closely align with the research objectives. These keywords were selected based on their relevance to wastewater management and encompassed a

Search Code

ITLE-ABS-KEY (("wastewater management*" OR "waste water management*" OR "sewage management*" OR "Industrial Waste Water*") AND (enabler* OR challenge* OR "Water Footprint Calculator" OR driver* OR "Water Calculator*") AND ("Waste Water Treatment*" OR "Water Treatment Plant" OR "Water Treatment Technology*") AND (sustainab* OR conservat* OR environment* OR protect*))

Fig. 5. Search code for scopus database.

wide spectrum of related concepts, such as treatment methods, pollution control, and environmental impact. The rationale behind each keyword choice was to capture a comprehensive range of studies that contribute to a holistic understanding of the field. By incorporating a well-structured and comprehensive search code, the review aimed to encompass a diverse array of publications that address various facets of industrial wastewater management.

3.3. Step-3

In order to critically appraise the publications data, necessary descriptive analysis have been performed by plotting corresponding graphs. It visualizes the literature data and is helpful to interpret the publishing behavior in the field of WWM, appropriately (ElMekawy et al., 2018). It allows identifying the trends and patterns in the data that may not be apparent through textual analysis alone. Moreover graphical analysis makes it easier to communicate the results of the study to others, as the information is presented in a clear and concise manner (refer Table 3). It can help in making informed decisions based on the data by providing a better understanding of the information. By using this new feature of Scopus data base, required graphical analysis of captured data has been performed. All the graphs are quoted in 1st column and their corresponding interpretation on 2nd column respectively.

3.4. Step-4

Literature classification through word mining, also known as co-occurrence analysis, is a technique used to identify patterns and relationships among words and terms in a large body of text (Wang et al., 2018). One commonly used tool for this type of analysis is the VOSviewer software, which can be used to create visualizations of co-occurring terms (Guo et al., 2019). To perform this type of analysis, a large corpus of text is typically gathered, such as a set of scientific articles or books on a particular topic (Chen et al., 2023). The text is then processed to identify the most common words and terms, and to analyze the frequency with which they appear in relation to each other (Tamala et al., 2022). VOSviewer software can be used to visualize the results of this analysis in the form of a network diagram (Oladinrin et al., 2023). Each term is represented as a node in the network, and the strength of the relationship between terms is represented by the thickness of the lines connecting them (Kim et al., 2021; Khan et al., 2023). By analyzing these visualizations, researchers can identify key themes and patterns within the corpus of text. Literature classification through word mining is commonly used in fields such as information science, where it can be used to identify important research trends and areas of focus (Fahimnia et al., 2015). It can also be used in other fields, such as marketing and advertising, to identify patterns in consumer behavior or preferences (, 2015 Zupic and Čater,).

The network visualization in VOSviewer describes items as labeled circles, where the size of the circle and label corresponds to the weight of the item (refer Fig. 6). Overlapping labels are avoided by not displaying some labels. The color of an item is determined by the cluster it belongs to. The lines between items represent links, with the 1000 strongest links displayed by default. The distance between items reflects the relatedness

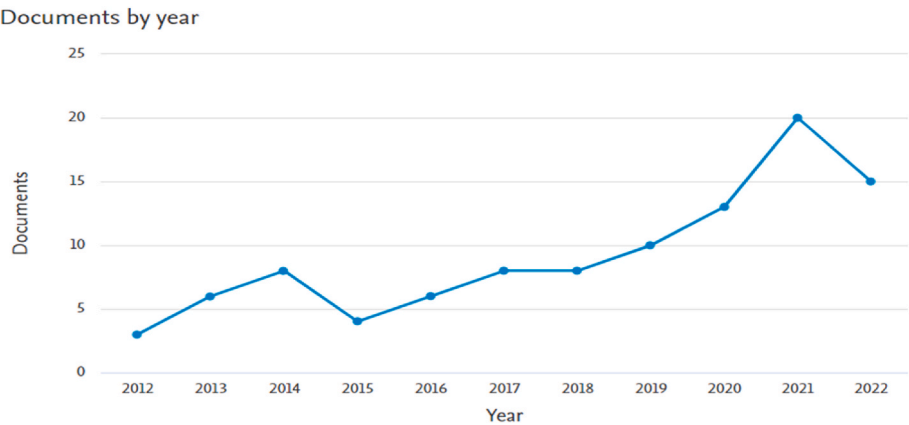
of the items in terms of co-citation links, with closer items indicating stronger relatedness. Fig. 6 provides an example of the network visualization, where the strongest co-citation links between journals are represented by lines.

Fig. 4 presents 22 terms with their respective occurrences and relevance scores. These highly repetitive terms can be the base (or criteria) for classifying the whole literature. The relevance score is a measure of how important a term is in the given context, which is likely related to wastewater treatment or management. The term “challenge” has the highest relevance score of 1.5955, indicating that it is the most important term in the context. Other highly relevant terms include “energy”, “wwtp”, “addition”, and “operation”. Terms such as “impact”, “study”, and “water” also have relatively high relevance scores. The network mapping also shows the number of occurrences for each term, with “challenge” having the highest occurrence of 89, followed by “wwtp” with 18 and “wastewater treatment plant” with 41. These terms are likely important in the given context and appear frequently in the literature. Overall, the network diagram provides valuable information on the importance and frequency of certain terms related to wastewater treatment or management, which can aid in understanding and analyzing relevant literature in this field. The terms like operation, study, water management and impact can be combined to generate drivers or enablers. At the end it can be said that the challenges and enablers are the hot terms, which seems to be the logical criteria to divide the literature related to WWM appropriately.

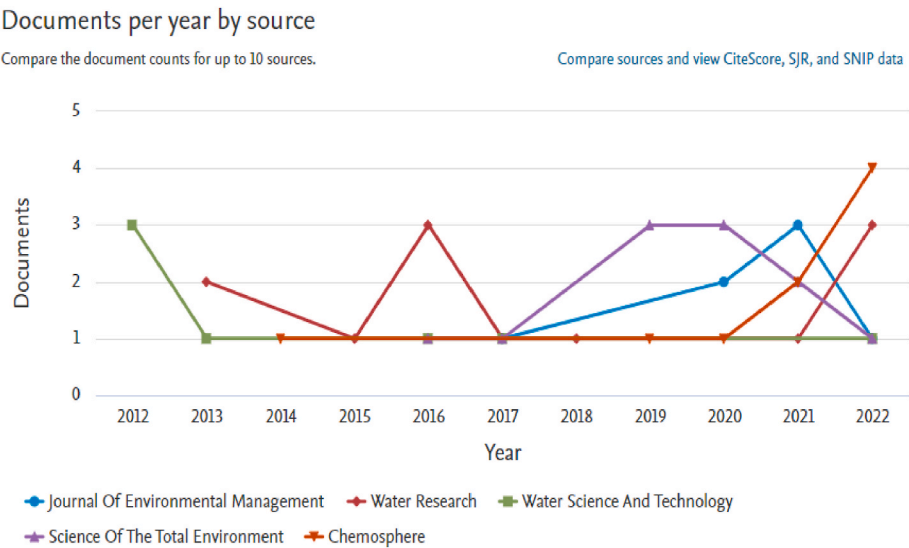
3.5. Step-5

Systematic Literature Reviews are a well-established method for identifying, evaluating and synthesizing evidence on a particular research question (Linnenluecke et al., 2020). One of the key steps in an SLR is ‘Screening Records (or relevant papers)’, which involves evaluating the title and abstract of each record to determine its relevancy to the designed research question. This process helps to identify records that are clearly irrelevant to the research question and eliminates them from further consideration, making the review more manageable and efficient (Ray et al., 2015). After screening, the remaining records are subjected to a full-text review to determine whether they meet the inclusion criteria for the review (Kitchenham et al., 2009). The screening process is critical in ensuring that the SLR is thorough and unbiased. However, it can also be time-consuming and resource-intensive, particularly for reviews that yield a large number of records. Various tools and techniques have been developed to aid in the screening process, such as by using machine learning algorithms to prioritize records (O’Mara Eves et al., 2015). In present case 101 papers have been scrutinized critically and found 22 of them irrelevant. These were found not to be significant with respect to present research objectives and hence ignored in present context to avoid any confusion or distraction. After this step, only 79 papers are left with to further carry out eligibility test. In the screening process, inclusion criteria were applied to ensure selected publications matched the research objectives in industrial wastewater management. The inclusion criteria included; relevance, subject alignment, document type, affiliations, keywords, global representation, publication year and stage. These ensured credible, pertinent records were chosen, allowing for a comprehensive analysis of

Table-3
Descriptive Analysis (101 studies).



The chart illustrates the annual production of documents between 2012 and 2022. There is a considerable fluctuation in the number of documents generated each year, with the highest being 20 in 2021 and the lowest being 3 in 2012. However, the overall trend over the last decade has been an increase in the number of documents produced. The majority of the documents (68 out of 103) were produced in the last five years (2018–2022), while the remaining 35 documents were produced in the seven years prior (2012–2017).



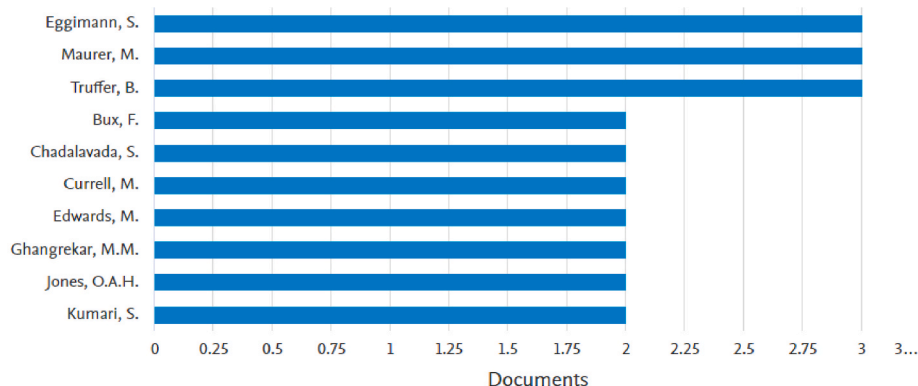
The top five Journals namely; Environmental Management, Water Research, Water Science and Technology, Chemosphere, and Science of the Total Environment have published varying numbers of documents each year from 2012 to 2023. The Journal of Environmental Management has published a consistently high number of documents every year, with a peak of 2384 documents in 2021. Water Research and Chemosphere have also shown a steady increase in the number of documents published over the years. On the other hand, Water Science and Technology has shown a declining trend in the number of documents published, with a steep drop in 2019 and a minimal number of publications in 2023. Science of the Total Environment has published a high number of documents in 2014 and 2015 but has not published any documents from 2016 onwards.

In summary, the analysis of the data shows that the Journal of Environmental Management, Water Research, and Chemosphere are leading sources in publishing documents on environmental management, water research, and chemistry. However, the number of publications by Water Science and Technology has declined over the years, and Science of the Total Environment has not published any documents recently.

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Table-3 (continued)**Documents by author**

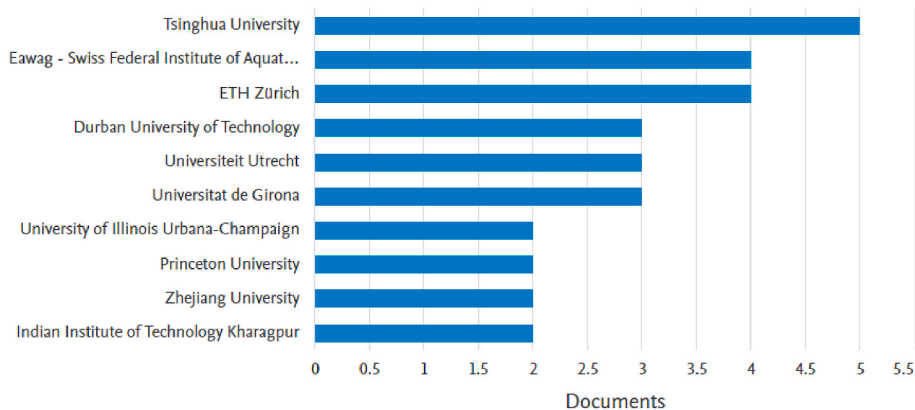
Compare the document counts for up to 15 authors.



The horizontal bar graph highlights the number of documents authored by top authors in the field of WWM during last decade. The graph plotted the top 10 authors in the field of WWM (out first 15). Eggimann, Maurer and Truffer having the highest number of documents at 3 each. Bux, Chadalavada, Currell, Edwards, Ghangrekar, Jones, Kumari, McCance, Pant, and Surapaneni each have 2 documents listed. Finally, there are two authors Aarestrup and Abatan, with only 1 document listed.

Documents by affiliation ⓘ

Compare the document counts for up to 15 affiliations.



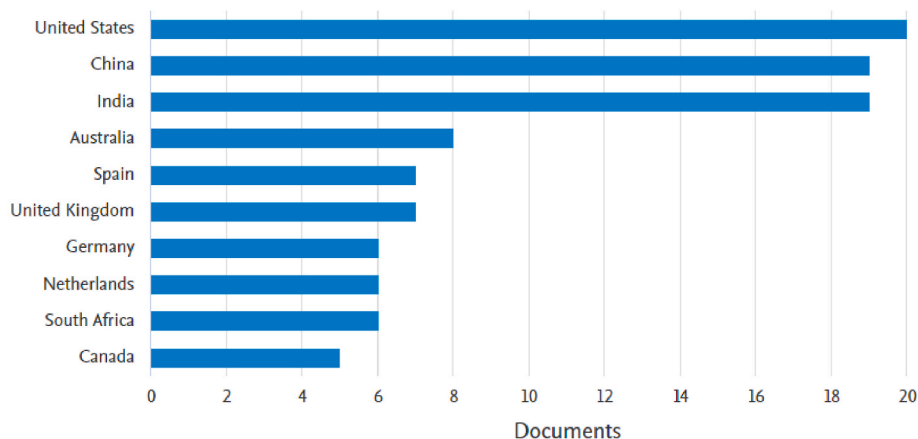
The bar graph delineates the number of documents affiliated with each institution in the given research area of WWM. Tsinghua University has the highest number of documents with 5 listed, followed by Eawag - Swiss Federal Institute of Aquatic Science and Technology and ETH Zürich, both with 4 documents listed. Durban University of Technology, Universiteit Utrecht, and Universitat de Girona each have 3 documents listed. There are several institutions with 2 documents listed, including University of Illinois Urbana-Champaign, Princeton University, Zhejiang University, Indian Institute of Technology Kharagpur, Opole University of Technology, Indian Institute of Technology Guwahati, The University of Newcastle, Australia, RMIT University, and Peking University. However, one possible observation is that there are a few institutions with a higher number of documents, which could indicate that they are more active or well-funded in this field or area.

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Table-3 (continued)

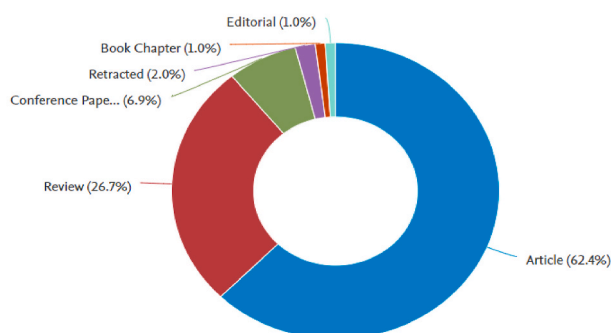
Documents by country or territory

Compare the document counts for up to 15 countries/territories.



The plot demonstrates the number of documents produced by each country/territory in the area of WWM. The United States, China, and India are the top three countries with the highest number of documents, all with 19 or 20 documents listed. Australia and Spain follow with 8 and 7 documents respectively, while the United Kingdom, Germany, and the Netherlands each have 6 documents listed. South Africa, Canada, France, and Belgium all have 4–6 documents listed, while Switzerland, Malaysia, and Mexico each have 3 documents listed. The top three countries with the highest number of documents are also the most populous countries in the world, which could indicate a correlation between population size and research output in this field or area.

Documents by type

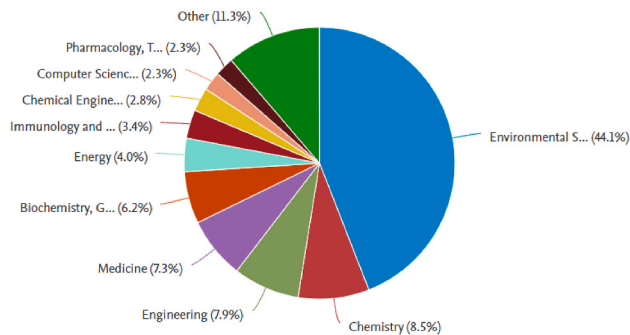


The pie chart defines the distribution of document types in the field of WWM during last ten years. Articles are the most common document type with 63 documents listed, followed by reviews with 28 documents listed. Conference papers are the third most common document type with 7 documents listed. There is one book chapter and one editorial listed. There are also two retracted documents listed, which are articles that have been withdrawn due to errors or misconduct. The articles are the most common document type, which could indicate that they are the primary mode of communication and dissemination of research in this field. The presence of retracted documents is also noteworthy, as it indicates a level of accountability and transparency in the research community. Further analysis and information about the documents and their content would be needed to make any meaningful conclusions.

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Table-3 (continued)

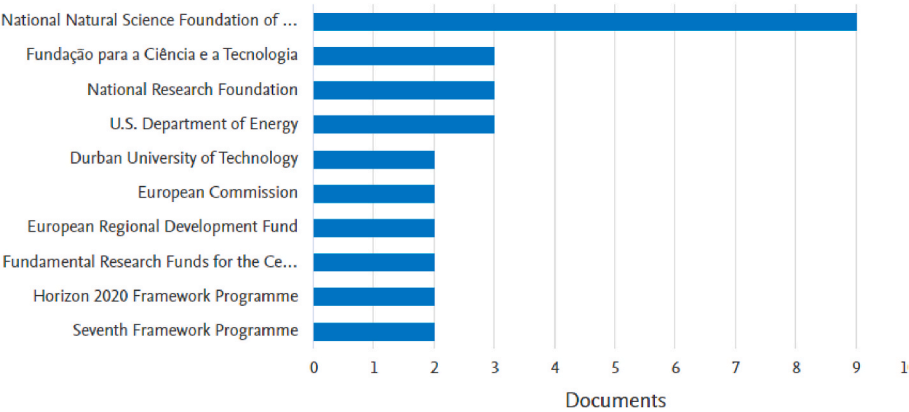
Documents by subject area



The pie chart represents the distribution of documents in different subject areas in the under study area of WWM. Environmental Science is the most represented subject area with 78 documents listed, followed by Chemistry with 15 documents, and Engineering with 14 documents. Medicine and Biochemistry, Genetics & Molecular Biology each have 13 and 11 documents listed respectively. Energy, Immunology & Microbiology, Chemical Engineering, Computer Science, Pharmacology, Toxicology, Pharmaceutics, Earth & Planetary Sciences, Economics, Econometrics, Finance, Materials Science, Physics & Astronomy, Agricultural & Biological Sciences, Mathematics and Social Sciences all have between 1 and 7 documents listed. The category of Multidisciplinary has only 1 document listed, while Neuroscience has only 1 document. Environmental Science is the most represented subject area, which could indicate a focus on the environmental impact or sustainability of the research in this field of WWM.

Documents by funding sponsor

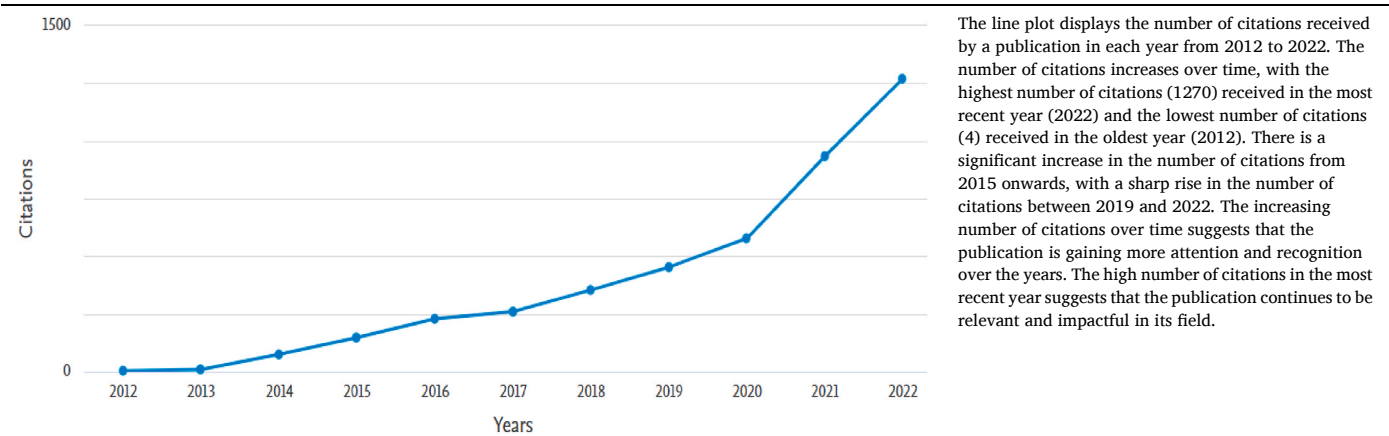
Compare the document counts for up to 15 funding sponsors.



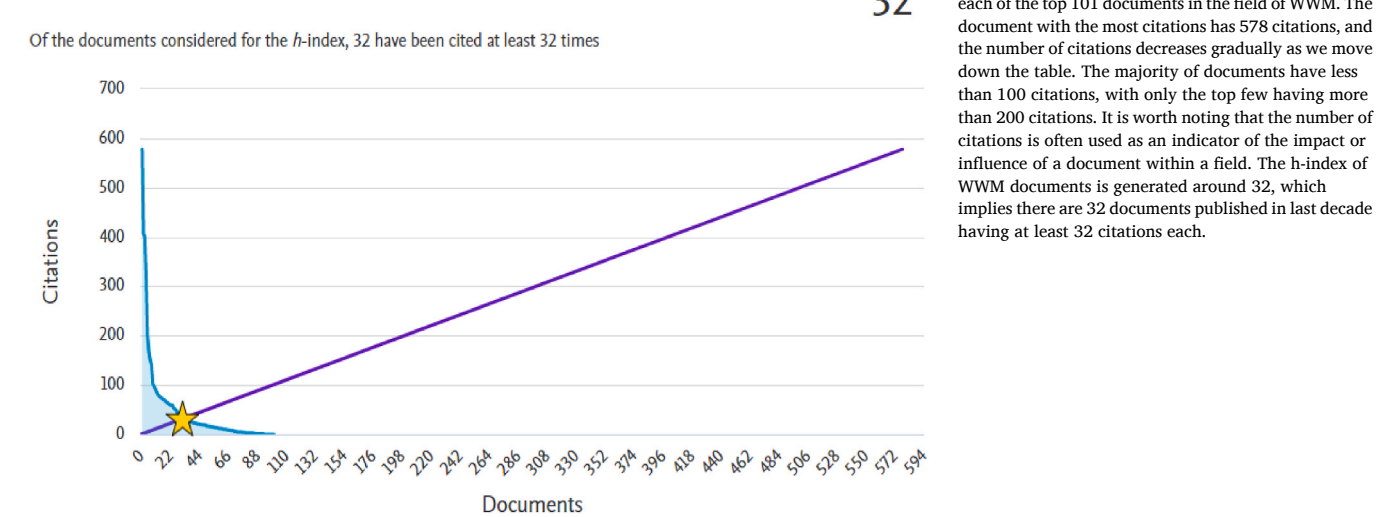
The horizontal bar graph depicts the distribution of documents based on their top 10 funding sponsors. The National Natural Science Foundation of China is the most frequent funding sponsor with 9 documents listed, followed by Fundação para a Ciência e a Tecnologia, National Research Foundation, and the U.S. Department of Energy, each with 3 documents listed. Several funding sponsors, including Durban University of Technology, European Commission, European Regional Development Fund and Fundamental Research Funds for the Central Universities, 2017 Framework Programme, and Seventh Framework Programme each have 2 documents listed. The research or studies in the field of WWM are highly dependent on funding from external sources, and that the National Natural Science Foundation of China is a significant funder in this area.

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Table-3 (continued)



These documents *h*-index



distribution, characteristics and trends in the field. Machine learning algorithms, specifically natural language processing (NLP), can also be employed alongside manual screening to categorize abstracts. NLP algorithms can train to identify relevant content patterns, resulting in quicker and more accurate abstract classification (Sodhi et al., 2022; Sheth et al., 2023). This collaborative approach will reduce time and resources, enhance screening efficiency and maintain accuracy in selecting pertinent abstracts.

3.6. Step-6

The eligibility of full-text articles during SLR is one of the significant steps (R van Dinter et al., 2021). This is done to ensure that the articles meet the inclusion criteria and have enough quality to be included in the final analysis. The process involves carefully reading the full text of each article and evaluating its relevance, validity and reliability (Leverett, 2014). The eligibility assessment is usually done by at least two independent reviewers to minimize bias and ensure consistency. Several studies have discussed the importance of conducting a thorough quality assessment of full-text articles during systematic literature reviews. For example, a study by Khan et al. (2018) emphasized the need to assess the

risk of bias in the included studies and recommended the use of standardized tools for quality assessment. Another study by Shea et al. (2007) highlighted the importance of evaluating the reporting quality of the studies to ensure that all relevant information is reported accurately. In the present study, screened 79 articles are checked for their eligibility by studying the full-length papers. Out of which 13 are realized as vague and were omitted from analysis. Full-text article assessment involved criteria such as applicability, valid methodology, reliable findings, peer review, citations, knowledge contribution, practical implications and diversity. Out of 13, 3 articles were removed because of poor methodology, 6 because of weak knowledge contribution and rest 4 were ignored because of irrelevancy. Rigorous evaluation ensured selected articles aligned with research goals and maintained quality standards in industrial wastewater management literature. In present SLR, the eligibility assessment involved two independent reviewers (co-authors) evaluating full-text articles. Disagreements were resolved through a consensus-building process, where reviewers discussed differing opinions and analyzed aspects of relevance, validity and reliability. In Around 11 articles, the consensus was not reached. So, a third senior-reviewer (the corresponding author) provided his input to make a final decision. This step ensured that the studies included (66

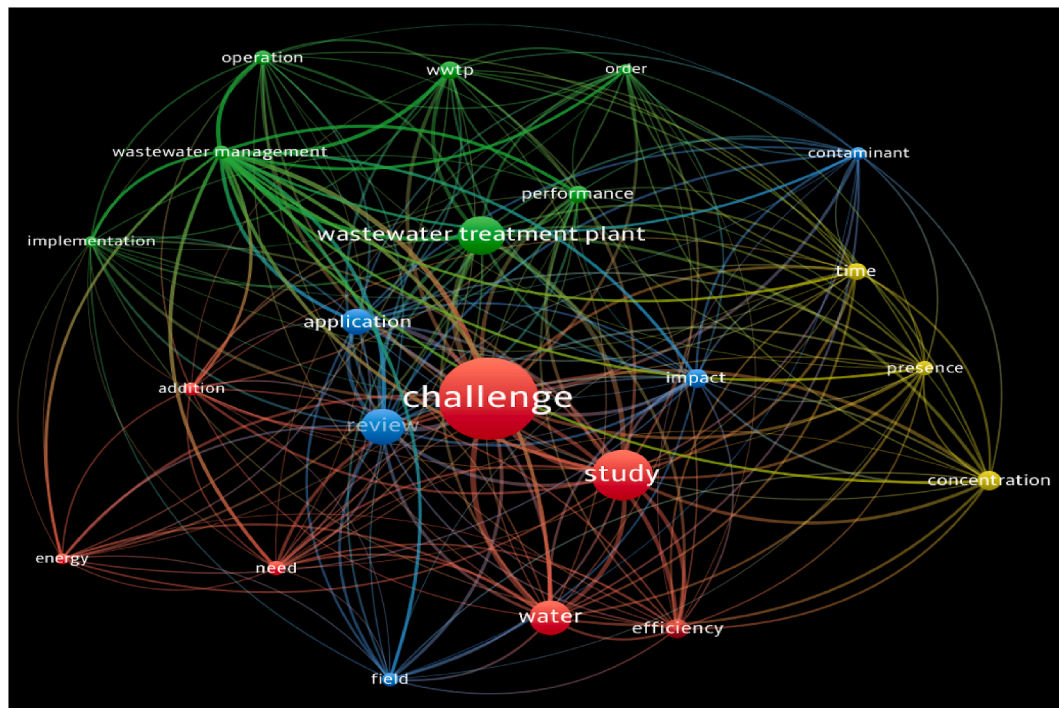


Fig. 6. Network map for literature classification.

manuscripts) in the review were of high quality, relevancy and strictly met the inclusion criteria. Conducting a thorough quality assessment also helped to reduce bias and increase the validity and reliability of the review.

3.7. Step-7

Now as per the predefined objectives, these 66 papers have been strictly scrutinized to skim-out the challenges and enablers in WWM. It has been found that 18 papers are written about wastewater and its other operational aspects, which are beyond the scope of present SLR. After neglecting these, only 48 studies have been explored to assess concerned challenges and enablers in the field of WWM. Necessary Qualitative Analysis was carried out as depicted in Table 4 ahead.

4. Discussions

This section of the review paper provides a comprehensive comparison and verifications of its findings with existing literature. Notable trends and patterns in industrial wastewater management are highlighted, drawing parallels with prior research. The review identifies common challenges and enablers in the field, aligning its observations with established studies. Several previous works have addressed challenges similar to those identified in this review. Ruane et al. (2008) discussed the impact of water scarcity on wastewater management, echoing the concern over limited water resources raised in our study. Similarly, Benzidia et al. (2021) emphasized the importance of sustainable treatment processes, aligning with our emphasis on the low sustainability of current treatment methods. This review finding on inadequate effluent discharge standards resonate with Jin et al.'s (2014) assertion of questionable WWTP effluent quality. Next, the emphasis on a circular bio-economy and resource recovery in this paper aligns with the approach advocated by Liu et al. (2021a,b) in their study of sustainable wastewater practices. Our recognition of the significance of data privacy and algorithmic development for deep learning in wastewater management aligns with the concerns raised by Ho and Goethals (2022) in their discussion of challenges in implementing AI

technologies.

In terms of research methodologies, the use of systematic literature review methodology parallels that of Karimidastenaie et al. (2021) in their exploration of water management challenges. Similarly, the integration of umbrella review and critical literature review, in this work finds resonance in the approach adopted by Wohlfart and Wagner (2023) in their comprehensive study of wastewater management enablers. In brief, this review paper's findings corroborate and extend existing knowledge in the field of industrial wastewater management. A study by Sindane and Modley (2022) conducted a similar review of challenges in wastewater management across different industries, corroborating our identification of common issues such as water scarcity, pollution, and inadequate treatment processes. Similarly, McQuaid et al. (2021) examined enablers for sustainable industrial wastewater management and highlighted the significance of effective policy support, echoing our emphasis on the importance of government regulations as a key enabler. To validate our findings regarding the role of technology, the research by Wawale et al. (2022) investigated the integration of AI and IoT in wastewater management, aligning with our recognition of emerging technologies as potential enablers. Moreover, the work of Verdaguer et al. (2010) demonstrated the relevance of multi-agent systems for complex urban wastewater management systems, reinforcing our identification of multi-agent systems as a challenge.

The focus on resource recovery and circular bio-economy aligns with the work of Liu et al. (2021a,b), which emphasized the importance of transitioning from a linear to a circular approach in wastewater management. Additionally, the concerns about the presence of microplastics and nanoparticles in water, highlighted in this review, are echoed in the study by Ferreira et al. (2016) affirming the observations of this SLR. The consistent alignment of the observations of this SLR with previous research accentuates the validity of the challenges and enablers it identified in industrial wastewater management. By harmonizing the results with established knowledge, this SLR has contributed to a more comprehensive understanding of the field and provides valuable insights for future research and practice.

Table-4

Qualitative analysis.

Sr. No.	Authors with Year of Publication	Research Area	Type of Industry	Technology used in WWM	Challenges	Enablers
1	Fu, G. et al. (2022).	Deep learning's role in urban water management: A review	Urban Cities	Deep Learning	Deep learning methods are still in their early stages of implementation within the water sector. Research challenges related to privacy of data, algorithmic development, reliability, multi-agent systems, and digital twins need to be addressed in order to apply deep learning effectively to urban waste water management.	Urban wastewater management requires accurate prediction of water demand, timely detection of leakage and contamination, effective evaluation of sewer defects, reliable prognosis of wastewater system status, regular monitoring of assets, and proper assessment of urban flooding risks for operational excellence.
2	Priyadarshini, M. et al. (2022).	Performance, advantages, and scale-up of emerging technologies in advanced oxidation processes	Textile industry of Italy	Advanced Oxidation Techniques	Sludge production and management in case of treatment plants of textile industry effluents. AOPs have high operating costs and reuses of catalytic materials are still not possible.	Among the emerging advanced oxidation processes, modified electrochemical oxidation, gamma radiation, and plasma-assisted systems show promising potential for sustainable wastewater treatment due to their environmental compatibility, safety, and ability to efficiently transform emerging contaminants. Scaling up trials and a thorough understanding of their reaction kinetics in complex wastewater matrices are essential for assessing the technical and economic feasibility of full-scale processes.
3	Santos, E. et al. (2022).	Economic assessment of energy consumption: Alternative nature-based technologies' applicability in wastewater treatment plants in Portugal	Portuguese WWTPs	Comparing WWT in terms of Energy Cost (algae based WTPs are best)	To improve corporate performance by making process economical and ecofriendly are another giant challenges to achieve high energy savings.	Algae-based wastewater treatment plants are emerging as a mandatory future solution for wastewater treatment, replacing traditional methods like activated sludge to reduce carbon footprint and operational costs. To ensure successful deployment of modern WWTPs, a comprehensive evaluation methodology is required, considering both qualitative aspects (e.g., technology type, wastewater volume and operating expenses) and quantitative aspects (e.g., specific energy consumption assessment, energy operating expenses, expenses per unit inhabitants).
4	James, A. (2022).	Ceramic-microbial fuel cell (C-MFC) for wastewater treatment: A mini review	WWTP in General	Using waste water as a feed for MFC (critical analysis)	To enable adaptability and commercialization of the MFC technology are some of the major bottlenecks	The future application of promising groundwater tracers will provide wastewater treatment plant (WWTP) operators and regulatory authorities with a more definitive tool for assessing groundwater contamination caused by site operations.
5	Baquero-Rodríguez, G. A. et al. (2022).	Technology selection in biological wastewater treatment: Impact of elevation	Municipal WWTPs	Narrative Review	Reduction of aeration costs and to assess the effect of elevation on the process of Municipal WWT are needed more insights.	The elevation along with temperature of the WWTP acts as a driving agents to reduce the aeration expenses in biological WWT of

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Table-4 (continued)

Sr. No.	Authors with Year of Publication	Research Area	Type of Industry	Technology used in WWM	Challenges	Enablers
6	Yu, S. et al. (2022).	Emerging nanomaterials in water purification and environmental remediation: MXenes	heavy metal ions, radionuclides and pollutants	MXenes, Nano materials	Challenges are toxicity and secondary contamination of nanomaterials while using MXene-based nanomaterials in the WWT.	municipal waste waters. The activated sludge process is influenced substantially by various environmental conditions like; temperature of atmosphere and its pressure along with relative humidity in different seasons at various Latitudes and heights. The MXenes are a group of materials that consist of transition metal nitrides, carbides or carbonitrides, which have excellent properties such as high conductivity, hydrophilicity, structural stability and sorption capacity due to their versatile surface chemistry. They can effectively remove pollutants based on their own physicochemical properties and the chemical nature of the pollutants. Moreover, MXene-based nanomaterials are highly efficient in wastewater management. Bismuth based visible light sensitive photocatalysts are non-toxic, are cheap and possess changeable configuration as well as excellent optical and chemical nature. These can be modified by doping, heterojunction, Z-scheme etc. and can inactivate bacteria. The ozone treatment was effective in eliminating the estrogenic potential of MPs, making it a suitable method for their removal. To prevent an increase in endocrine loading in the water body after heavy rainfall, point sources such as rain overflow basins can be set up upstream of WWTPs. In addition, EBMs have been successfully used for compliance monitoring of estrogenic compounds. Not Discussed
7	Subhiksha, V. et al. (2022).	Recent advances in bismuth-based photocatalysts for organic pollutant degradation in aqueous solutions: A review	WWTPs in dye, paper and pulp, textiles, tannery and bleaching, pharmaceuticals etc.	Advanced oxidation processes (AOPs) by photocatalysts	Pristine BBP has limitations such as poor charge carrier mobility, inadequate bandgap, limited charge separation ability and limited light absorption capability. Additionally, carrier trapping, carrier recombination and thermal instability are also challenges.	The utilization of solar energy in photocatalysis processes, solar thermal electrochemical processes, solar desalination of brackish water, and solar advanced oxidation processes has been demonstrated to be cost-effective and promising for the future of wastewater treatment.
8	Wolf, Y. et al. (2022).	Improvement of water and wastewater quality: Comprehensive analysis of endocrine potential using effect-based methods in a full-scale ozonation plant	WWTP Aachen-Soers, the largest full-scale ozonation plant in Europe.	Ozone treatment, ERα- and AR CALUX assays making ROB	WWTPs release micropollutants (MPs) into nearby water bodies, particularly endocrine disrupting compounds (EDCs), when they discharge treated wastewater.	
9	Gudiukaite, R. et al. (2021).	Bioprocesses for bioenergy and value-added products recovery from wastewater: A review	Bio processes	Recovery of Bio-Energy from WWTPs	Worldwide, wastewater and activated sludge pose significant challenges. These arise from their production by all industries, laundries and human settlements, among others.	
10	Pandey, A. K. et al. (2021).	Solar energy utilization in wastewater treatment: Challenges and research trends	Solar Powered WWTPs	Rising Solar Technology	Energy storage systems pose a challenge for WWTPs due to the increase in overall expenses. Additionally, the need for immobilized catalysts and the intermittent nature of solar power are other challenges.	

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Table-4 (continued)

Sr. No.	Authors with Year of Publication	Research Area	Type of Industry	Technology used in WWM	Challenges	Enablers
11	Colella, M. et al. (2021).	Challenges and opportunities for efficient water use and circular wastewater management: The case of Campania Region, Italy	WWM in Campania Region (Italy)	CBE processes	It is necessary to adopt circular bioeconomy (CBE) processes for WW management. CBE processes involve the use of closed systems to promote the systematic recycling, reuse, and recovery of bioproducts and by-products, as well as the reduction of waste generation.	The adoption of sustainable practices in wastewater management is seen as a pathway towards achieving a circular bioeconomy (CBE). Reuse of wastewater can reduce the volume and stress on water bodies while also allowing for the recovery of valuable nutrients and bioproducts.
12	Lyu, Y. et al. (2021).	Managing water sustainability in textile industry: Adaptive multiple stakeholder collaboration approach	Textile Industries in China	Adaptive Multiple Stakeholder Collaboration (AMSC)	Textile industry wastewater treatment involves intensive chemical consumption and can result in waterborne pollution.	Industrial clustering can improve the efficiency of wastewater treatment for industries, such as textiles, through the implementation of a two-stage treatment system comprising of both in-plant and centralized treatment facilities.
13	Saravanan, A. et al. (2021).	Effective methodologies for toxic pollutant removal in water/wastewater treatment: Processes and applications for sustainable development	Pollutants	Audit Review Approach	Human practices pollute water and contaminate water with heavy metals, dyes, pathogenic and organic compounds. Therefore the demand for pollutant free water has risen which requires water pollution control.	Membrane separation and adsorption techniques have been found to be more effective in controlling water pollution compared to conventional methods and established recovery processes. Additionally, diverse wastewater treatment technologies with an emphasis on pre-treatment and post-treatment processes are driving innovation in wastewater management.
14	Moussavi, S. et al. (2021).	Environmental impacts of construction and operations: Relative life cycle assessment of small mechanical wastewater treatment plants	WWTPs in Small Communities	Assessing the environmental impacts of construction and operations of WWTPs in small communities.	Rural communities find it hard to construct and operate WWTPs	Scenario sensitivity analyses have revealed that longer design lives result in reduced environmental impact during construction, and end-of-life considerations do not significantly affect the environmental sustainability of a wastewater treatment plant (WWTP).
15	McCance, W. et al. (2021).	Decoupling wastewater impacts from hydrogeochemical trends in groundwater resources	WWTP in South-Eastern Australia	Isotopic analysis	Hydrogeochemical plotting tools, multivariate statistics, and targeted isotopic analysis need to be periodically conducted to improve the characterization of important hydrogeochemical influences and anthropogenic inputs.	Long-term analysis of major ions and contaminants in wastewater, routinely collected by WWTP operators, can be analyzed using hydrogeochemical plotting tools, multivariate statistics, and targeted isotopic analysis to better understand hydrogeochemical influences and anthropogenic inputs.
16	Abbasi, N. et al. (2021).	Quality and cost analysis of a wastewater treatment plant using GPS-X and CapdetWorks simulation programs	Designing and construction	GPS-X software and CapdetWork software simulation using the Zargandeh treatment plant data (Tehran)	Technical and economic simulation and modeling are crucial for the design, construction, and prediction of wastewater treatment plant requirements. Wastewater management is a major challenge that must consider human health, environmental regulations, wastewater treatment plant costs, and water reusability.	CapdetWork software was used to calculate the economic costs of the wastewater treatment project, including implementation, maintenance, and energy usage. Contact stabilization was determined to be the more cost-effective option.
17	Karthigadevi, G. et al. (2021).	Chemico-nanotreatment methods for persistent organic pollutants and xenobiotics removal in water: A review	Chemical Industry	Nano-technology	Persistent pollutants can accumulate in the environment and cause bioaccumulation. To address the human health risks posed	Not Discussed

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Table-4 (continued)

Sr. No.	Authors with Year of Publication	Research Area	Type of Industry	Technology used in WWM	Challenges	Enablers
18	Cong, W. et al. (2021).	Wastewater governance in textile industrial parks in China	Binhai textile industrial park, china	event sequence analysis, an emerging polycentric governance model	by xenobiotics such as endocrine disrupting compounds (EDCs) and personal care products (PCPs), methods for their removal need to be developed. However, these methods have limitations due to microbial contamination. Sometimes government support becomes a challenge to address wastewater problems in textile industries.	The new model of wastewater treatment focuses on involving multiple parties to enhance cost-effectiveness and address compliance issues. Three conditions must be met to transition to this new model, including government support, implementing monitoring systems, and developing the capacity of actors involved. Since the effectiveness of current water treatment in eliminating SARS-CoV-2 is not confirmed, multiple treatment barriers should be used to ensure complete removal of the virus from water. However, additional scientific research is needed to develop effective viral disinfection strategies for wastewater treatment. Strategies for investigating wastewater treatment during viral pandemics could include regular testing of disinfectants, taking into account various factors that influence viral survival, evaluating performance in different environmental conditions, considering the impact of wastewater quality, and assessing the environmental effects of disinfection technology. Fungal laccases have the ability to convert micropollutants in municipal wastewater, which can reduce toxicity in harsh conditions, making them ideal biocatalysts for bioremediation.
19	Kataki, S. et al. (2021).	Wastewater treatment strategies during COVID-19 pandemic	Any Industry	COVID Impact	Not Discussed	Fungal laccases have the ability to convert micropollutants in municipal wastewater, which can reduce toxicity in harsh conditions, making them ideal biocatalysts for bioremediation. Fuzzy logic evaluation is an effective method for intelligent control of the aerobic stage in wastewater treatment. Soft-sensors aided by AI provide accurate real-time and offline estimates of difficult-to-measure parameters in wastewater treatment plants, while single and hybrid AI methods can be used to estimate pollutant concentrations and develop monitoring and early-warning systems for aquatic and atmospheric environments. Additionally, AI modeling can be used to predict short-term, mid-
20	Spina, F. et al. (2020).	Laccases treatment for municipal wastewater micropollutants	Municipal waste Italy	Bioremediation using fungal laccase enzyme	Micropollutants can be problematic in WWM. The most prevalent micropollutants identified were bis (2-ethylhexyl) phthalate, diethyl phthalate, and ketoprofen.	
21	Wang, Y. et al. (2020).	Fate modelling of micropollutants in water systems	AI in WWM	Single and hybrid AI methods	The features, benefits, and constraints of AI techniques, both singular and hybrid, for addressing WWM are summarized and disregarded.	

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Table-4 (continued)

Sr. No.	Authors with Year of Publication	Research Area	Type of Industry	Technology used in WWM	Challenges	Enablers
22	Magwaza, S. T. et al. (2020).	Hydroponic systems for urban wastewater treatment and vegetable production	Hydroponics & Munciple WWM	Hydroponics	Insufficient sanitation and wastewater disposal facilities can lead to environmental and public health issues, such as water scarcity, nutrient-depleted soils, and pollution in urban areas.	term, and long-term solid waste generation, and various artificial neural networks can be applied to solid waste recycling and reduction, which can improve the capabilities of wastewater treatment plants. Combining hydroponic systems with municipal wastewater treatment can decrease costs related to removing pollutants, as well as reducing the maintenance and energy requirements of traditional wastewater treatment methods. By using the treated wastewater for crop production, the system allows for the continual use of wastewater while simultaneously removing pollutants and nutrients (such as nitrogen and phosphorus), resulting in improved food security and environmental protection. This also helps reduce health risks to farmers, crops, and consumers that might result from exposure to wastewater.
23	Qu, J. et al. (2019).	Municipal wastewater treatment in China	Munciple waste China	Mini-review	The inadequate development of sewers and sludge disposal facilities, the limited sustainability of treatment processes, uncertain WWTP effluent discharge standards, and the lack of global considerations for balancing wastewater management, human society, and the environment are all issues.	The development of modern WWTPs aims to transform them from facilities solely focused on removing pollutants to ones that also generate energy, recover water and fertilizer resources, and contribute to the overall urban ecosystem.
24	Isosaari, P. et al. (2019).	Ionic liquid-based water treatment for organic pollutants	Organic Industries	Critical Review	Choosing non-hazardous ionic liquids, developing appropriate technological applications, meeting high testing standards for specific uses, and scaling up the entire process of removing, disposing, and regenerating ionic liquids for pollutant removal are all significant challenges. Furthermore, the toxicity evaluation of water-soluble ionic liquids and their cost implications are major concerns.	Not Discussed
25	Chiramba, R. et al. (2019).	Activated carbon from poultry feathers for wastewater treatment	Industry discharging lead ions to water at Lake Chivero, Zimbabwe	Activated Carbon	Water bodies in Zimbabwe are facing the presence of heavy metal ions, such as lead and chromium, which can have negative environmental and socio-economic consequences. Treatment can be expensive and may necessitate the use of chemicals and activated carbon.	Over five million tonnes of feathers are produced each year from poultry consumption. By incorporating feathers in a suitable manner, the effectiveness of biological reactions can be improved by increasing the concentration of the adsorbent and the duration of contact.

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Table-4 (continued)

Sr. No.	Authors with Year of Publication	Research Area	Type of Industry	Technology used in WWM	Challenges	Enablers
26	Jafarinejad and Jiang (2019).	Wastewater treatment for petroleum refineries and petrochemical plants	Petroleum refineries and petrochemical plants, Iran	Systematic Literature Survey	Water scarcity in Middle East areas along with limited wastewater management creates survival situations for refineries.	There are new hybrid technologies that can be developed to enhance the capacity of wastewater treatment, upgrade the quality of discharged water, and cope with the sudden release of dangerous materials and organic matter. These technologies also promote the reduction of water usage, the reuse of treated wastewater, and the mitigation of environmental impacts.
27	Enfrin, M. et al. (2019).	Nano/microplastics in water and wastewater treatment	Industry disposing Nano/microplastics in water	Comprehensive review	Nano and microplastics in water have become a significant environmental issue. It is difficult to detect and understand their fate due to limited analytical techniques available. Because of their small size and diverse chemical properties, nano and microplastics can move easily through water and wastewater treatment processes, posing challenges for operational and process stability.	Not Discussed
28	Guo, Z. et al. (2019).	Green energy integration for municipal wastewater treatment plants	Municipal wastewater treatment plants	Narrative Review	A considerable amount of energy is necessary to meet wastewater discharge standards.	The effectiveness of water treatment using IoT and green chemistry technologies has been demonstrated.
29	Wang, B. et al. (2019).	Derivatization for acid screening in oilfield refinery wastewater	Oil refinery, North China	Monitoring the behavior of Recalcitrant Chlorinated Naphthenic Acids	The untargeted scanning chemical profiling approach has the potential to identify unknown pollutants or biological markers, but identifying the structures of unknown compounds remains a significant challenge.	The approach of non-targeted scanning chemical profiling has a high potential for detecting unknown pollutants or new biological indicators.
30	Sancho, I. et al. (2019).	Carbon redirection in wastewater treatment plants	Industry producing Organic Waste	High-rate activated sludge (HRAS)	High-rate activated sludge (HRAS) is a technology that converts organic compounds to energy as biogas. However, there is still a long way to go in maximizing energy recovery from the organic load.	The development of new configurations for wastewater treatment plants is driven by legislation that requires stricter effluent requirements and the desire to make plants energy self-sufficient or even energy-positive.
31	McCance, W. et al. (2018).	Emerging contaminants as groundwater tracers for wastewater impacts in urban areas	Urban and pre-urban areas	Narrative Review	Groundwater contamination due to wastewater leakage or infiltration is a significant concern, especially from inorganic nutrients and persistent organic compounds. The assessment of ions, nutrients, bacteria, and conventional tracers cannot always accurately identify potential sources of contamination.	In the future, groundwater tracers that show promise could be applied to give wastewater treatment plant operators and regulatory authorities a more definitive way to assess contamination of groundwater associated with the plant's operations.
32	Mullins, D. et al. (2018).	Image processing for turbidity measurement in wastewater	Domestic and Dairy units	Automation	Automation for measuring effluent turbidity lacks easy operation and requires frequent human intervention and maintenance.	An automated system that provides a multipoint analysis for measuring the turbidity of large volumes of wastewater effluent is available and can be easily replicated.

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Table-4 (continued)

Sr. No.	Authors with Year of Publication	Research Area	Type of Industry	Technology used in WWM	Challenges	Enablers
33	Chen, Y. et al. (2017).	Microbial community regulation in wastewater treatment systems	Chemical Industry	Critical review on microbial contributions	Recent studies have improved our understanding of the relationship between microbial community structure and wastewater treatment process performance. However, challenges remain in fully utilizing microbial information, optimizing microbial community structure, and maintaining process performance in wastewater treatment systems.	Not Discussed
34	Eggimann, S. et al. (2016).	Cost of hybrid wastewater systems	Regional WWTPs	Framework for specifying minimum cost-connection rates	Determining the best connection rate for regional wastewater treatment is a recent challenge that has caught the attention of academics and professionals worldwide.	Not Discussed
35	Zhang, Q. H. et al. (2016)	Urban wastewater treatment in China	Urban Cities, China	Narrative Review	The challenge of selecting appropriate WW Treatment Technologies is still lingering along with effective administrative measures.	Not Discussed
36	Debnath et al. (2016)	Congo red-ferric chloride interaction in wastewater treatment	Chemical Processes	Thermodynamic aspect of WWTs	Purification of industrial water having anionic dyes is a gigantic challenge.	Chemical interaction of FeCl ₃ with Congo red (CR) dye has been found very much significant for removal of the dye from industrial waste water. It is a very quick and cost effective chemical method for removal of anionic dye (CR)
37	Wicklein et al. (2016)	Modelling best practices for wastewater treatment plants using computational fluid dynamics	Any Industry	CFD Modeling	Not Discussed	The use of computational fluid dynamics (CFD) modelling is expanding in the field of wastewater treatment (WWT) and is being applied to address more intricate issues.
38	Gaonkar and Kande (2014)	Automation systems for water and wastewater applications	Automatic WWTPs	Automation in WWTPs	Objective of nations worldwide is to check water wastage and contamination as water is the most abundant, most used and most depleted resource today. Human activities like inefficient agriculture and unsustainable water usage have led to water scarcity, unauthorized water use, and reduced crop yields. The ageing infrastructure in drinking water distribution networks has led to unmonitored leakage and uncontrolled water flow.	Information and Communication Technologies (ICTs) can be used as these are cost effective and require less human efforts and can be applied to domestic and industrial fronts.
39	Bayart et al. (2014)	Water Impact Index for water foot printing	Any Industry	Water footprint calculator	Improved water management is a rising sustainability challenge with climate change.	The wastewater treatment plant helps decrease the Water Impact Index of the wastewater management system. The paper proposes a simple method for water foot printing that takes into account volume, scarcity, and quality issues, which is a starting point for improving our comprehension and evaluation of the environmental effects of human activities on water resources.

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Table-4 (continued)

Sr. No.	Authors with Year of Publication	Research Area	Type of Industry	Technology used in WWM	Challenges	Enablers
40	Chaplin (2014)	Electrochemical advanced oxidation processes for water treatment	EAOPs	Critical Review	EAOP technologies face challenges due to toxic byproduct formation and low electro-active surface areas.	
41	Neale et al. (2013)	Engineered nanomaterials in wastewater treatment plants	Industry having Nano waste	Conventional Review	The presence of ENM surface coating and organic matter/surfactants poses a major threat.	Combining separation techniques with methods to evaluate particle size and chemical makeup seems to be the best strategy for handling wastewater. For a more comprehensive comprehension of the destiny and influence of Engineered Nanomaterials (ENM) connected with WWTPs, more accurate experimental designs with better measurement of ENM characteristics under the chosen test settings are needed.
42	Warner et al. (2013)	Shale gas wastewater disposal impacts on water quality in Pennsylvania	Oil refinery, USA	Empirical Assessment	Safe disposal of liquid waste from US oil and gas production is a major challenge due to large volumes and high levels of contaminants.	Not Discussed
43	Somlyódy and Patziger (2012)	Urban wastewater development in Europe	20 LARGE WWT plants in Europe located in six countries (Bulgaria, the Czech Republic, Hungary, Poland, Romania and Slovakia).	Comprehensive data collection	The nutrients are wasted along with waste water. It needs upgrading options and abundant amount of Energy to avoid nutritional wastages.	The factors that drive the water industry, such as GDP, water pricing, funding, and laws, should be carefully examined and analyzed.
44	Garrido-Baserba et al. (2012)	Knowledge management for integrated assessment of WWTP configurations during design	Design and Construction Industry	Knowledge management approach	Designing integrated plant layout for new wastewater projects has become challenging due to the increasing number of innovative technologies, stricter regulations, and new methodological approaches.	The integration of expert interviews, legislation, specialized literature, and engineering experience into the knowledge bases is one of the main achievements of this study. This methodology is presented as an essential tool that generates customized Process Flow Diagrams (PFD) tailored to each specific case, taking into account the process unit interactions and user-specified requirements and objectives.
45	Dong et al. (2012)	Model for assessing renovation scheme in integrated urban drainage system	Redesigning of Urban WWTPs, China	Integrated system	Due to urban development, redesigning drainage systems has become necessary, leading to new challenges for WWTPs.	Not Discussed
46	Martin Ruel et al. (2012)	Fate of substances in wastewater treatment plants with regards to Water Framework Directive	Municipal wastewater treatment lines	Role of guidelines and future legislations	Detecting and removing micropollutants at low concentrations is a challenge. Conventional activated sludge plants can remove 80% of the load, but two-thirds of the removed substances end up in sludge. Common micropollutants include Cd, DEHP, diuron, alkylphenols, and chloroform.	Not Discussed

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Table-4 (continued)

Sr. No.	Authors with Year of Publication	Research Area	Type of Industry	Technology used in WWM	Challenges	Enablers
47	Corominas et al. (2013)	Life cycle assessment for wastewater treatment: current state of the art	LCA of any Industry	comprehensive review	Not Discussed	The text emphasizes the importance of ISO standards to maintain quality and transparency in the application of Life Cycle Assessment (LCA) methods in Wastewater Treatment (WWT). The challenges that arise in LCA applications in WWT are also discussed, such as the need to shift from pollutant removal to resource recovery, adaptation to new target compounds, and the development of regional factors. Additionally, the text highlights the need to improve the quality of data and reduce uncertainty, as well as better integration and communication with decision-makers.
48	Plósz et al. (2013)	Modelling micro-pollutant fate in wastewater collection and treatment systems	Fate of Micropollutants in any Industry	comprehensive review	Not Discussed	To better anticipate the elimination of contaminants in centralized treatment plants, it is recommended to supplement traditional, heuristic, and other fate models with more mechanistic approaches, given the significant advances in monitoring and detecting MPs (microplastics) in wastewater.

5. Conclusions

In conclusion, this systematic review sheds light on the intricate landscape of industrial wastewater management. Through an in-depth analysis of challenges and enablers, this study has featured the critical importance of addressing environmental concerns while optimizing resource utilization. By identifying key regulatory, technological and operational factors, this review paves the way for sustainable solutions that can enhance both economic viability and ecological resilience. This synthesis serves as a comprehensive resource for stakeholders seeking to navigate the complexities of industrial wastewater management, fostering a path towards effective, responsible, and holistic practices in the industrial sector. Furthermore, it highlights the need for interdisciplinary collaboration among researchers, policymakers, industries and communities to tackle the multifaceted issues associated with industrial wastewater. As industries continue to evolve and expand, it is imperative to adopt innovative strategies that minimize pollution, promote circular economy principles and ensure compliance with stringent regulations.

The challenges discussed, such as water scarcity, pollution control and sustainable resource management, highlight the urgency of adopting proactive measures. Leveraging emerging technologies, process optimization, and efficient treatment methodologies can significantly alleviate the negative impacts of industrial wastewater discharge on ecosystems and public health. While challenges are significant, the enablers identified present promising opportunities for progress. Policy support, stakeholder engagement, technological advancements and sustainable practices collectively form a roadmap towards a more responsible and efficient industrial wastewater management paradigm. It is essential for industries to prioritize the integration of environmentally conscious practices, not only to meet regulatory requirements but

also to contribute positively to the overall well-being of societies and the environment. In conclusion, this systematic review acts as a catalyst for informed decision-making in industrial wastewater management. By comprehensively evaluating challenges and enablers, we aim to inspire a transformative shift towards responsible and sustainable practices, fostering a harmonious coexistence between industries and the environment.

The fluctuations in the annual number of documents produced in the field of water, wastewater and environmental management carry important implications. These variations could be attributed to several factors. Shifts in research focus, driven by emerging environmental concerns or technological advancements, may lead to increased attention in certain years. Policy changes or regulations can also drive research activity, causing spikes in document production when new guidelines are implemented or revised. Conversely, decreases in output may result from periods of relatively stable policy frameworks or mature research topics. Advancements in technology, such as new treatment methods or monitoring techniques, can influence the direction and intensity of research, contributing to annual variations. The implications of these fluctuations highlight the dynamic nature of the field, as well as the interplay between scientific progress, policy dynamics and societal needs, all of which shape the trajectory of research output over time. Moreover, the publication trends of specific wastewater management (WWM) journals offer insights into research shifts and gaps. Declining publications in “Water Science and Technology” suggest changing research focus or competition. Absence of recent articles in “Science of the Total Environment” might indicate unexplored themes or methods. Such trends impact visibility and information diffusion in the WWM community. Investigating and addressing these trends can lead to valuable research directions aligned with societal needs. The decline prompts exploration of new frontiers, while understanding the absence

unveils opportunities for interdisciplinary approaches. In summary, these trends shape WWM discourse, directing research towards vital themes and innovative perspectives.

Additionally, the relevance score and network mapping of terms are powerful tools that have greatly aided in identifying essential research themes and trends in wastewater treatment and management. These techniques involve advanced data analysis and visualization to uncover hidden patterns and relationships within a large body of research literature. Through this analysis, specific insights can be gained.

- **Emerging Themes:** High relevance scores and dense clusters in network maps can indicate emerging research themes. For instance, a sudden increase in the relevance score of terms related to “nano-technology” or “sustainable practices” might signify growing interest in these areas.
- **Shifts in Research Focus:** Changes in the relevance scores and connections between terms can reflect shifts in research focus over time. For example, a decline in the relevance score of older wastewater treatment methods might indicate a shift towards newer, more efficient technologies.
- **Interdisciplinary Connections:** Network mapping can reveal interdisciplinary connections between terms that are not immediately obvious. Identifying these connections can lead to innovative collaborations between researchers from different fields.
- **Research Gaps:** Sparse clusters in network maps may indicate areas that are underexplored. These gaps present opportunities for researchers to delve deeper into specific topics that have not received adequate attention.
- **Policy and Decision-Making:** The insights gained from these analyses can inform policy decisions and resource allocation. For instance, if a term related to “policy implementation” emerges as highly relevant, it suggests a need for research that informs effective policy formulation.
- **Technology Adoption:** If certain terms related to specific wastewater treatment technologies consistently show high relevance scores and strong connections, it may indicate a trend towards the adoption and exploration of these technologies.

Overall, the relevance score and network mapping provide a comprehensive overview of the research landscape, identifying trends, shifts, gaps, and interdisciplinary connections. These insights are valuable for shaping future research directions, guiding funding decisions, prioritizing research topics and facilitating collaborations. By leveraging these techniques, stakeholders in wastewater treatment and management can make informed decisions to advance the field and address critical challenges effectively.

One of the specific outputs of this paper is an explanation of literature classification through word mining and co-occurrence analysis. It introduces the concept of using tools like VOSviewer software to create visualizations of co-occurring terms in a large body of text. The text explains the process of gathering a corpus of text, identifying common words and terms, analyzing their frequency and relationships, and visualizing the results as a network diagram. The output also describes how the network visualization in VOSviewer works, including the representation of items as labeled circles, the use of colors to indicate clusters, the representation of links between items, and the use of distance to reflect relatedness. The provided text illustrates the application of this technique in fields like information science and marketing to identify trends and patterns. The term “challenge” has the highest relevance score, indicating its significance. Other relevant terms include “energy,” “wwtp,” “addition,” “operation,” “impact,” “study,” and “water.” The network mapping also displays term occurrences, with “challenge” having the highest occurrence, followed by “wwtp” and “wastewater treatment plant.” The text highlights that these terms are crucial and frequently appear in the literature. It emphasizes that the network diagram provides valuable insights into term importance and

frequency, aiding the understanding and analysis of relevant literature in wastewater treatment or management. Additionally, the text suggests the possibility of combining terms to generate drivers or enablers and identifies challenges and enablers as important criteria to classify literature in the field.

5.1. Mapping of the literature findings with research questions

The systematic review has captured following findings as per pre-drafted research questions.

5.1.1. Challenges

The utilization of deep learning techniques in the water industry is relatively new, and there are currently five obstacles in applying deep learning to urban wastewater management: safeguarding data privacy, improving algorithmic development, ensuring dependability, dealing with multi-agent systems, and creating digital twins. The treatment of sludge in textile industry effluent plants presents challenges, and advanced oxidation processes (AOPs) have high operating costs and limited reuse of catalytic materials. There is a need to improve corporate performance by making processes economical and eco-friendly, and to enable adaptability and commercialization of the microbial fuel cell (MFC) technology. Other challenges include reducing aeration costs, assessing the effect of elevation on the municipal wastewater treatment process, and addressing the toxicity and secondary contamination of nanomaterials when using MXene-based nanomaterials in wastewater treatment.

Pristine BBP has several limitations, including insufficient light absorption, poor mobility of charge carriers, low charge separation ability, inadequate bandgap, carrier trapping and recombination, and thermal instability. Wastewater treatment plants discharge micro-pollutants, mainly EDCs, which pose a threat to the environment and human health. The construction and operation of wastewater treatment plants are difficult for rural communities. There are various methods for addressing the removal of xenobiotics to prevent human health risks, but microbial contamination can limit their effectiveness. The lack of government support is sometimes a challenge in addressing wastewater problems in textile industries. The presence of micro-plastics and nanoparticles in water is a significant environmental issue, and their detection is still a challenge due to inadequate analytical techniques.

Wastewater management faces several challenges, including water scarcity, nutrient depletion in soil, and urban pollution, which can lead to environmental and public health issues. Other challenges include the inadequate development of sewer systems and sludge disposal facilities, the sustainability of treatment processes, the lack of global coordination in harmonious development between wastewater management, human society & nature and questionable effluent discharge standards from wastewater treatment plants. In addition, high testing requirements for individual uses, scaling up of the entire pollutant removal and disposal process using non-hazardous ionic liquids, toxicity assessment of water-soluble ionic liquids, and their cost are all challenges.

Heavy metal ions such as lead and chromium in Zimbabwe’s water bodies are also significant problems, and treatment can be expensive and require chemicals and activated carbon. In the Middle East, water scarcity and limited wastewater management create survival situations for refineries. Groundwater contamination due to the infiltration of wastewater, especially with inorganic nutrients like ammonia and nitrate as well as persistent organic compounds, is also a major threat. Automated turbidity measurement of effluents with minimal human intervention is necessary. The non-targeted scanning chemical profiling approach shows potential to identify unknown pollutants or novel biological markers, but identifying unknown compounds is still challenging. Despite the use of high-rate activated sludge (HRAS) technology to redirect carbon towards energy as biogas, maximizing energy recovery from the organic load remains a challenge.

5.1.2. Enablers

To operate urban wastewater management effectively, several factors need consideration, such as predicting water demand, detecting contamination and leakage, evaluating sewer defects, monitoring assets, assessing urban flooding and predicting the state of wastewater systems. Modified electrochemical oxidation, gamma radiation and plasma-assisted systems are the most promising methods for sustainable and successful implementation in wastewater treatment due to their compatibility, environmental safety and efficient conversion of contaminants. Biotechnology, specifically algal wastewater treatment plants, is a mandatory future solution for cost-effective wastewater treatment with reduced carbon footprint. Other effective methods for wastewater management include groundwater tracers, MXene-based nanomaterial, bismuth-based visible light-sensitive photo-catalysts and effect-based techniques.

The future of wastewater management is expected to shift towards a circular bio-economy (CBE), where wastewater is reused to recover nutrient elements and bio-products, reducing the volume and pressure on water bodies. Industrial clustering can also be used for more effective wastewater treatment, while membrane separation and adsorption methodologies are more efficient than traditional methods. To achieve a new model of multi-actor participation, policy supports from the government, monitoring systems and capacity building for actors are required. Additionally, extra precautions must be taken to remove viruses like SARS-CoV-2 from water. Fungal laccases, fuzzy logic and AI-aided soft-sensors are performing excellently for online/offline estimation of hard-to-measure parameters in wastewater treatment plants. AI modeling of short-term, mid-term and long-term solid waste generation can be used to design monitoring and early-warning systems for both aquatic and atmospheric environments.

These insights regarding challenges and enablers can have several positive outcomes.

- **Enhanced Treatment Efficiency:** Addressing challenges and leveraging enablers can result in more effective and efficient wastewater treatment processes, leading to improved water quality and reduced environmental impacts.
- **Sustainable Solutions:** Overcoming challenges related to cost, energy consumption, and resource utilization can lead to more sustainable and economically viable wastewater management practices.
- **Environmental Protection:** Successful implementation of enablers can lead to reduced contamination, protection of ecosystems, and decreased threat to human health from pollutants and micro-pollutants.
- **Policy Formulation:** The identified challenges and enablers can guide policymakers in developing effective regulations and strategies to improve wastewater treatment practices, ensuring long-term environmental sustainability.
- **Research Direction:** Researchers can focus on addressing specific challenges and harnessing enablers to drive innovation and advancements in wastewater treatment technologies and methodologies.

So, these challenges and enablers provide valuable insights that can guide decision-makers, researchers, and industry professionals to make informed choices, implement effective solutions, and drive positive impacts in the field of wastewater treatment and management.

The identified challenges and enablers in wastewater management have far-reaching implications for environmental management and sustainable development. By addressing challenges like pollution, resource inefficiency and public health risks, the field aligns with Sustainable Development Goals (SDGs) related to clean water, energy, innovation and responsible consumption. Embracing enablers such as advanced technologies and circular economy approaches these goals, promoting resource conservation, public health improvement and scientific advancement. These findings mark the interconnection between

wastewater management and broader sustainability objectives, emphasizing the need for interdisciplinary collaboration and innovative solutions to ensure environmental health, economic prosperity and social well-being. Next, the identified challenges and enablers in wastewater management have profound implications for improving practices, addressing environmental issues and enhancing public health. Overcoming deep learning obstacles can lead to data-driven decision-making and optimized processes. Resolving sludge treatment complexities ensures cleaner water bodies and safer communities. Economic and eco-friendly processes align with circular economy principles. Microbial fuel cell commercialization contributes to sustainable energy generation. Addressing emerging contaminants and nanomaterials safeguards water quality and public health. Integrated wastewater management and advanced technologies enhance asset management and pollution prevention. Circular bio-economy practices reduce waste and resource consumption. AI and advanced tools optimize treatment efficiency. These efforts collectively advance environmental management and sustainable development goals.

5.2. Limitations

This systematic review has following limitations that should be considered. Firstly, the publication source is Scopus database only. Other databases like Web of Sciences, Science Direct, PubMed etc. are not involved. Secondly, the scope of this review is also limited and inculcates publications from the previous decade only. Thirdly, the review may not account for contextual factors such as regional or cultural differences that can affect the implementation and effectiveness of wastewater management practices. The review is limited by its methodology, such as the search strategy or inclusion criteria, which can impact the comprehensiveness of the review.

Moreover, there are several limitations of not considering meta-analysis while doing review. Without meta-analysis, a review may have limited statistical power and may not be able to draw definitive conclusions. Similarly without conducting meta-analysis, it may be difficult to identify the sources of heterogeneity between studies. This can make it challenging to draw meaningful conclusions about treatment effects. A review may not be able to provide a comprehensive and generalizable summary of the available evidence.

This study has not discussed the water footprint and its calculation. A water footprint is a measure of the total amount of water used to produce a product or service, including both direct and indirect water use. By not considering water footprint and its calculations, the review may fail to provide a comprehensive understanding of the water-related environmental impact of industrial activities. Likewise, the water footprint approach can provide insights into how water is being used within an industry and identify opportunities to reduce water consumption and wastewater generation. This can lead to more sustainable water management practices that not only benefit the environment but also reduce costs and increase efficiency for businesses. Therefore, the omission of water footprint and its calculation in the review could limit its applicability in addressing the water-related sustainability challenges faced by industries. Incorporating the concept of water footprint and its calculation can provide a more complete picture of the environmental impact of industrial activities and guide the development of effective wastewater management strategies.

5.3. Future scope

There are several future plans that can help cope up with the gaps in wastewater management. The implementation of advanced wastewater treatment technologies can help in improving the quality of treated wastewater, thereby reducing the risk of waterborne diseases. Advanced treatment technologies will play a pivotal role in addressing complex contaminants and achieving higher purification standards in industrial wastewater treatment. Notable technologies include Advanced

Oxidation Processes (AOPs), which use powerful oxidants to break down pollutants; Membrane Bioreactors (MBRs), combining biological treatment with membrane filtration for high-quality effluent; Electrocoagulation (EC), inducing coagulation via electrical current; Constructed Wetlands, using aquatic plants and microbes for treatment; Membrane Filtration, employing porous membranes for selective removal; Adsorption Technologies, utilizing materials to bind pollutants; and Zero Liquid Discharge (ZLD), recovering and reusing water while minimizing waste. These technologies have proven successful in diverse industries. Membrane filtration ensures pharmaceutical water purity, EC removes heavy metals in electronics manufacturing, and AOPs treat dye-laden textile effluents. These innovations not only enhance wastewater management practices but also address environmental concerns and improve public health, aligning with the broader landscape of sustainable development.

Decentralized wastewater treatment systems can be an effective solution for areas that lack centralized wastewater treatment infrastructure. These systems can treat wastewater at the source, reducing the burden on centralized treatment facilities. Integration of green infrastructure, such as constructed wetlands and green roofs, can help in reducing the quantity and improving the quality of wastewater. Increased public awareness can play a critical role in reducing the generation of wastewater at the source. This can be achieved through educational campaigns and community engagement programs. Successful public awareness campaigns and community engagement programs, like "Save Water, Save Life" in California and "Love Your River" in the UK, promote wastewater reduction. Barriers such as lack of awareness and disconnection can be overcome through interactive workshops, user-friendly apps, and collaborative efforts. Utilizing social media to showcase success stories and positive impacts can also motivate participation. Overcoming these barriers empowers communities to actively contribute to wastewater reduction, fostering sustainability and environmental health.

Improved policy and regulatory framework can help in promoting sustainable wastewater management practices. This can be achieved through the development of appropriate standards, regulations and incentives for the implementation of sustainable wastewater management practices. Specific policy measures and regulations can play a vital role in driving sustainable wastewater management practices. Strict effluent standards, financial incentives, and penalties for non-compliance motivate industries to adopt cleaner technologies. Water conservation pricing, reuse mandates, and transparency enhance responsible practices in communities. Collaborative partnerships and capacity-building programs promote innovation and effective implementation. These measures align economic interests with environmental responsibility, fostering adoption of sustainable wastewater practices and contributing to a healthier ecosystem. Investment in research and development can help in the development of new and innovative wastewater treatment technologies, which can improve the efficiency and sustainability of wastewater management practices. Breakthroughs are anticipated in areas like nanotechnology, bioremediation and energy-efficient processes. Nanomaterials show potential for efficient pollutant removal due to their high surface area and reactivity. Bioremediation techniques, including microbial fuel cells and algal bioreactors, offer sustainable treatment options. Energy-efficient processes like anaerobic digestion and electrochemical systems reduce environmental impact. Ongoing projects such as "Nano-remediation for Industrial Wastewater Treatment" and "Microbial Fuel Cells for Bioelectricity Generation from Wastewater" showcase promising developments. Additionally, innovations like solar-driven photocatalysis and membrane-based separation methods hold potential. These advancements can revolutionize wastewater treatment, enhancing efficiency, sustainability, and environmental protection. At end, Meta-analysis along with bibliometric review is recommended while carrying out systematic reviews, because it provides a quantitative synthesis of the results from multiple studies. A systematic review aims to identify, select, and critically evaluate all

relevant studies on a specific research question, but it does not provide a statistical summary of the findings. Suitable Meta-analysis and bibliometric analysis must be planned in future to draw more robust findings in the area of WWM.

The concept of water footprint or water footprint calculator is gaining popularity in the field of wastewater management as it helps to assess the amount of water used by an individual or an organization and to identify opportunities for reducing water consumption. In the future, it is expected that the use of water footprint calculators will become more widespread, and that this will be used not only by businesses and industries, but also by individuals and households. This will enable people to understand their own water usage patterns and to take steps to reduce their water consumption. The water footprint concept quantifies the total volume of freshwater used in the production and consumption of goods and services, including direct water use and the indirect water embedded in products. It provides insights into the global water consumption patterns and aids in sustainable water management. For wastewater management, understanding the water footprint helps identify areas of high water consumption and pollution, enabling more efficient use and treatment. Industries can identify water-intensive processes and implement water-saving technologies to reduce their water footprint. Analyzing water footprints throughout the supply chain helps in choosing suppliers with lower water impacts. Similarly, designing products with lower water footprints can lead to more sustainable manufacturing. Identifying water-intensive and polluting processes helps in targeting areas for improved wastewater treatment. Like assessing the water footprint of textile production highlights high water usage, driving efforts to reduce consumption and improve wastewater treatment. Likewise, by calculating the water footprint of agricultural products, water-intensive crops can be grown in suitable regions, reducing the overall water footprint. But, calculating water footprints involves various factors, such as geographic location, climate and local water availability, making it challenging to account for all variables accurately. Addressing the water footprint alone might overlook other environmental concerns like energy consumption, land use and pollution. Obtaining accurate data for water consumption across entire supply chains can be difficult, leading to potential inaccuracies in footprint calculations. In nut shell, the water footprint concept empowers industries to make informed decisions for more responsible water management, which can lead to reduced water consumption, minimized pollution and improved wastewater treatment practices.

Another important aspect of the future of wastewater management is the use of innovative technologies to treat and recycle wastewater. Technologies such as reverse osmosis, membrane bioreactors and electrochemical treatment systems are already being used to treat wastewater, but there is still scope for further innovation in this area. Also, regulations around wastewater management will become more stringent in the future, with a greater emphasis on reducing the environmental impact of wastewater discharge. This will require businesses and industries to invest in wastewater treatment and management technologies, as well as to adopt more sustainable practices that reduce water usage.

At end, the future of wastewater management will involve a combination of technological innovation, regulatory compliance and individual & collective action to reduce water consumption and minimize the environmental impact of wastewater discharge. The use of water footprint calculators is one of the key enablers in achieving this goal.

This modified systematic literature review provides insights into the current state of industrial wastewater management and suggests avenues for future research and practice. It emphasizes the need for holistic and collaborative approaches that involve all relevant stakeholders and consider the economic, environmental and social dimensions of wastewater management. This approach ensures cost-effective strategies, minimizes environmental impact, enhances community well-being and aligns with Sustainable Development Goals. Collaborative efforts harness diverse expertise and lead to innovative solutions. Examples

include integrated urban water management and circular economy models. Challenges include balancing priorities and implementing supportive policies. Ultimately, holistic approaches enhance resilience, address interconnected challenges and contribute to a more sustainable and resilient wastewater management system.

Declaration

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Annexure-1: Refine Filters (Research Limited to)

Number of results: 101				Scopus Database Accessed on Nov 17, 2022							
OPEN ACCESS	Nos	YEAR	Nos. of Publications	DOCUMENT TYPE	Nos	SOURCE TITLE	Nos	COUNTRY	Nos	SOURCE TYPE	Nos
All Open Access	35	2022	15	Article	67	Science Of The Total Environment	14	China	24	Journal	98
Gold	7	2021	16	Review	28	Water Science And Technology	13	United States	24	Conference Proceeding	3
Hybrid Gold	10	2020	18	Conference Paper	3	Water Research	10	India	13		
Bronze	6	2019	14	Note	3	Journal Of Environmental Management	9	Australia	11		
Green	20	2018	5			Chemosphere	8	United Kingdom	10		
		2017	5			Environmental Science And Technology	6	Germany	6		
		2016	7			Environment International	6	Poland	6		
		2015	2			Water Science And Technology Water Supply	3	Canada	4		
		2014	6			Environmental Monitoring And Assessment	2	France	4		
		2013	5			Environmental Pollution	2	Italy	4		
		2012	8			Environmental Research	2	Netherlands	4		
						Resources Conservation And Recycling	2	Spain	4		
						Advanced Science Letters	1	Switzerland	4		
						Bioresource Technology	1	Belgium	3		
						Critical Reviews In Environmental Science And Technology	1	South Africa	3		
						Energies	1	South Korea	3		
						Environmental Management	1	Sweden	3		
						Environmental Science And Policy	1	Denmark	2		
						Environmental Sciences Processes And Impacts	1	Ireland	2		
						Environmental Toxicology And Chemistry	1	Lithuania	2		
						Frontiers Of Environmental Science And Engineering	1	Malaysia	2		
						International Journal Of Environmental Research And Public Health	1	Portugal	2		
						International Journal Of Life Cycle Assessment	1	Saudi Arabia	2		
						Iop Conference Series Earth And Environmental Science	1	Taiwan	2		
						Journal Of Environmental Chemical Engineering	1	Turkey	2		
						Journal Of Hazardous Materials	1	Argentina	1		
						Journal Of The Air And Waste Management Association	1	Botswana	1		
						Journal Of Water And Health	1	Colombia	1		
						Matec Web Of Conferences	1	Czech Republic	1		
						Npj Clean Water	1	Finland	1		
						Proceedings Of The IEEE International Conference On Industrial Technology	1	Greece	1		
						Water Air And Soil Pollution	1	Hungary	1		
						Water Environment Research	1	Indonesia	1		
						Water Practice And Technology	1	Iran	1		
						Water Research X	1	Lebanon	1		
						Water Switzerland	1	Mexico	1		
								Norway	1		

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(continued)

Number of results: 101				Scopus Database Accessed on Nov 17, 2022							
OPEN ACCESS	Nos	YEAR	Nos. of Publications	DOCUMENT TYPE	Nos	SOURCE TITLE	Nos	COUNTRY	Nos	SOURCE TYPE	Nos
								Qatar	1		
								Russian Federation	1		
								Singapore	1		
								Tanzania	1		
								Thailand	1		
								Viet Nam	1		
								Zambia	1		
								Zimbabwe	1		

References

- Abbas, N., Ahmadi, M., Naseri, M., 2021. Quality and cost analysis of a wastewater treatment plant using GPS-X and CapdetWorks simulation programs. *J. Environ. Manag.* 284, 111993 <https://doi.org/10.1016/j.jenvman.2021.111993>.
- Al-Tohamy, R., Ali, S.S., Li, F., Okasha, K.M., Mahmoud, Y.A.G., Elsamahy, T., Jiao, H., Fu, Y., Sun, J., 2022. A critical review on the treatment of dye-containing wastewater: ecotoxicological and health concerns of textile dyes and possible remediation approaches for environmental safety. *Ecotoxicol. Environ. Saf.* 231, 113160 <https://doi.org/10.1016/j.ecoenv.2021.113160>.
- Amador, J.A., Göres, J.H., Loomis, G.W., Lancellotti, B.V., 2018. Nitrogen loading from onsite wastewater treatment systems in the greater narragansett bay (Rhode Island, USA) watershed: magnitude and reduction strategies. *Water Air Soil Pollut.* 229 (3), 65. <https://doi.org/10.1007/s11270-018-3714-4>.
- Atangana, E., Oberholster, P.J., Turton, A.R., 2021. Will the extraction of COVID-19 from wastewater help flatten the curve? *Chemosphere* 271, 129429. <https://doi.org/10.1016/j.chemosphere.2020.129429>.
- Baquero-Rodríguez, G.A., Martínez, S., Acuña, J., Nolasco, D., Rosso, D., 2022. How elevation dictates technology selection in biological wastewater treatment. *J. Environ. Manag.* 307, 114588 <https://doi.org/10.1016/j.jenvman.2022.114588>.
- Barton, H., Berbel-Filho, W.M., Consuegra, S., Francis, L., Tizaoui, C., Conlan, R.S., Teixeira, S.R., 2018. Ultrasensitive environmental assessment of xeno-estrogens in water samples using label-free graphene immunosensors. *Anal. Biochem.* 548, 102–108. <https://doi.org/10.1016/j.chemosphere.2020.129429>.
- Bayart, J.-B., Worbe, S., Grimaud, J., Aoustin, E., 2014. The Water Impact Index: a simplified single-indicator approach for water footprinting. *Int. J. Life Cycle Assess.* 19 (6), 1336–1344. <https://doi.org/10.1007/s11367-014-0732-3>.
- Benzdia, S., Makaoui, N., Bentahar, O., 2021. The impact of big data analytics and artificial intelligence on green supply chain process integration and hospital environmental performance. *Technol. Forecast. Soc. Change* 165, 120557. <https://doi.org/10.1016/j.techfore.2020.120557>. Get rights and content.
- Birtok, R.I., Karpuzcu, M.E., Ozturk, I., 2022. Occurrence of priority substances in urban wastewaters of Istanbul and the estimation of the associated risks in the effluents. *Environ. Monit. Assess.* 194 (6), 426. <https://doi.org/10.1007/s10661-022-09840-w>.
- Boguniewicz-Zablocka, J., Klosok-Bazan, I., Capodaglio, A.G., Ploskonka, J., 2018. Planning the optimal solution for wastewater management in rural areas - case study. *MATEC Web of Conferences* 174, 1035. <https://doi.org/10.1051/mateconf/201817401035>.
- Burge, S.R., Hristovski, K.D., Burge, R.G., Hoffman, D.A., Saboe, D., Chao, P., Taylor, E., Koenigsberg, S.S., 2020. Microbial potentiometric sensor: a new approach to longstanding challenges. *Sci. Total Environ.* 742, 140528 <https://doi.org/10.1016/j.scitotenv.2020.140528>.
- Cao, W., Yang, Q., 2020. Online sequential extreme learning machine based adaptive control for wastewater treatment plant. *Neurocomputing* 408, 169–175. <https://doi.org/10.1016/j.neucom.2019.05.109>.
- Chai, W.S., Tan, W.G., Halimatul Munawaroh, H.S., Gupta, V.K., Ho, S.-H., Show, P.L., 2021. Multifaceted roles of microalgae in the application of wastewater biotreatment: a review. *Environ. Pollut.* 269, 116236 <https://doi.org/10.1016/j.envpol.2020.116236>.
- Chaplin, B.P., 2014. Critical review of electrochemical advanced oxidation processes for water treatment applications. *Environ. Sci. J. Integr. Environ. Res.: Process. Impacts* 16 (6), 1182–1203. <https://doi.org/10.1039/c3em00679d>.
- Chen, Y., Lan, S., Wang, L., Dong, S., Zhou, H., Tan, Z., Li, X., 2017. A review: driving factors and regulation strategies of microbial community structure and dynamics in wastewater treatment systems. *Chemosphere* 174, 173–182. <https://doi.org/10.1016/j.chemosphere.2017.01.129>.
- Chen, M., Yao, T., Wang, K., 2023. The economic impact of climate change: a bibliometric analysis of research hotspots and trends. *Environ. Sci. Pollut. Control Ser.* 30 (16), 47935–47955. <https://doi.org/10.1007/s11356-023-25721-2>.
- Chiramba, R., Charis, G., Fungura, N., Danha, G., Mamvura, T., 2019. Production of activated carbon from poultry feathers for waste water treatment. *Water Sci. Technol.* 80 (8), 1407–1412. <https://doi.org/10.2166/wst.2019.388>.
- Colella, M., Ripa, M., Coccozza, A., Panfilo, C., Ulgiati, S., 2021a. Challenges and opportunities for more efficient water use and circular wastewater management. The case of Campania Region, Italy. *J. Environ. Manag.* 297, 113171 <https://doi.org/10.1016/j.jenvman.2021.113171>.
- Compton, M., Willis, S., Rezaie, B., Humes, K., 2018. Food processing industry energy and water consumption in the Pacific northwest. *Innovat. Food Sci. Emerg. Technol.* 47, 371–383. <https://doi.org/10.1016/j.ifset.2018.04.001>.
- Cong, W., Li, X., Qian, Y., Shi, L., 2021. Polycentric approach of wastewater governance in textile industrial parks: case study of local governance innovation in China. *J. Environ. Manag.* 280, 111730 <https://doi.org/10.1016/j.jenvman.2020.111730>.
- Corominas, L., Foley, J., Guest, J.S., Hospido, A., Larsen, H.F., Morera, S., Shaw, A., 2013. Life cycle assessment applied to wastewater treatment: state of the art. *Water Res.* 47 (15), 5480–5492. <https://doi.org/10.1016/j.watres.2013.06.049>.
- Crow, J., Murray, J.A., 2020. Online distance learning in biomedical sciences: community, belonging and presence. *Biomedical Visualisation* 6, 165–178. https://doi.org/10.1007/978-3-030-37639-0_10.
- Debnath, A., Thakur, A., Bhowmik, K.L., Bera, A., Saha, B., 2016. Thermodynamic studies on the interaction of Congo red with ferric chloride in aqueous medium for waste water treatment. *Adv. Sci. Lett.* 22 (1), 242–245. <https://doi.org/10.1166/asl.2016.6774>.
- Dong, X., Zeng, S., Chen, J., Zhao, D., 2012. An integrated urban drainage system model for assessing renovation scheme. *Water Sci. Technol.* 65 (10), 1781–1788. <https://doi.org/10.2166/wst.2012.076>.
- Dwivedi, K.A., Huang, S.-J., Wang, C.-T., 2022. Integration of various technology-based approaches for enhancing the performance of microbial fuel cell technology: a review. *Chemosphere* 287, 132248. <https://doi.org/10.1016/j.chemosphere.2021.132248>.
- Efeovbokhan, V.E., Olurotimi, O.O., Yusuf, E.O., Abatan, O.G., Alagbe, E.E., 2019. Production of clay filters for waste water treatment. *December J. Phys. Conf.* 1378 (3), 032028. <https://doi.org/10.1088/1742-6596/1378/3/032028>. IOP Publishing.
- Eggimann, S., Truffer, B., Maurer, M., 2016. The cost of hybrid waste water systems: a systematic framework for specifying minimum cost-connection rates. *Water Res.* 103, 472–484. <https://doi.org/10.1016/j.watres.2016.07.062>.
- ElMekawy, A., Hegab, H.M., Pant, D., Saint, C.P., 2018. Bio-analytical applications of microbial fuel cell-based biosensors for onsite water quality monitoring. *J. Appl. Microbiol.* 124 (1), 302–313. <https://doi.org/10.1111/jam.13631>.
- Enfrin, M., Dumée, L.F., Lee, J., 2019. Nano/microplastics in water and wastewater treatment processes – origin, impact and potential solutions. *Water Res.* 161, 621–638. <https://doi.org/10.1016/j.watres.2019.06.049>.
- Fahimnia, B., Sarkis, J., Davarzani,

- during conceptual design. *Water Sci. Technol.* 66 (1), 165–172. <https://doi.org/10.2166/wst.2012.087>.
- Ge, S., Usack, J., Ma, B., 2018. Advances and challenges at the waste-to-bioenergy/biorefinery nexus. *BioMed Res. Int.* <https://doi.org/10.1155/2018/3642363>, 2018.
- Gerbersdorf, S.U., Cimattoribus, C., Class, H., Engesser, K.-H., Helbich, S., Hollert, H., Lange, C., Kranert, M., Metzger, J., Nowak, W., Seiler, T.-B., Steger, K., Steinmetz, H., Wieprecht, S., 2015. Anthropogenic Trace Compounds (ATCs) in aquatic habitats - research needs on sources, fate, detection and toxicity to ensure timely elimination strategies and risk management. *Environ. Int.* 79, 85–105. <https://doi.org/10.1016/j.envint.2015.03.011>.
- Gingerich, D.B., Sun, X., Behrer, A.P., Azevedo, I.L., Mauter, M.S., 2017. Spatially resolved air-water emissions tradeoffs improve regulatory impact analyses for electricity generation. *Proc. Natl. Acad. Sci. USA* 114 (8), 1862–1867. <https://doi.org/10.1073/pnas.1616071114>.
- Gudiukaite, R., Nadda, A.K., Gricajeva, A., Shanmugam, S., Nguyen, D.D., Lam, S.S., 2021. Bioprocesses for the recovery of bioenergy and value-added products from wastewater: a review. *J. Environ. Manag.* 300, 113831 <https://doi.org/10.1016/j.jenvman.2021.113831>.
- Guldhe, A., Kumari, S., Ramanna, L., Ramsundar, P., Singh, P., Rawat, I., Bux, F., 2017. Prospects, recent advancements and challenges of different wastewater streams for microalgal cultivation. *J. Environ. Manag.* 203, 299–315. <https://doi.org/10.1016/j.jenvman.2017.08.012>.
- Gulfam-E-Jannat, S., Golui, D., Islam, S., Saha, B., Rahman, S.M., Bezbaruah, A.N., Iskander, S.M., 2023. Industrial Water Demand and Wastewater Generation: Challenges for Bangladesh's Water Industry. *ACS ES&T Water*. <https://doi.org/10.1021/acsestwater.3c00023>.
- Guo, Z., Sun, Y., Pan, S.-Y., Chiang, P.-C., 2019. Integration of green energy and advanced energy-efficient technologies for municipal wastewater treatment plants. *Int. J. Environ. Res. Publ. Health* 16 (7), 1282. <https://doi.org/10.3390/ijerph16071282>.
- Harahap, J., Gunawan, T., Suprayogi, S., Widyastuti, M., 2021. A review: domestic wastewater management system in Indonesia. *IOP Conf. Ser. Earth Environ. Sci.* 739 (1), 012031 <https://doi.org/10.1088/1755-1315/739/1/012031>.
- Harzing, A.W., 2013. A preliminary test of Google Scholar as a source for citation data: a longitudinal study of Nobel prize winners. *Scientometrics* 94 (3), 1057–1075. <https://doi.org/10.1007/s11192-012-0777-7>.
- Hjelmsø, M.H., Hellmér, M., Fernandez-Cassi, X., Timoneda, N., Lukjancenko, O., Seidel, M., Elsäßer, D., Aarestrup, F.M., Löfström, C., Bofill-Mas, S., Abril, J.F., Girones, R., Schultz, A.C., 2017. Evaluation of methods for the concentration and extraction of viruses from sewage in the context of metagenomic sequencing. *PLoS One* 12 (1), e0170199. <https://doi.org/10.1371/journal.pone.0170199>.
- Ho, L., Goethals, P., 2022. Machine learning applications in river research: trends, opportunities and challenges. *Methods Ecol. Evol.* 13 (11), 2603–2621. <https://doi.org/10.1111/2041-210X.13992>.
- Ioannidis, J.P., 2016. The mass production of redundant, misleading, and conflicted systematic reviews and meta-analyses. *Milbank Q.* 94 (3), 485–514. <https://doi.org/10.1111/1468-0009.12210>.
- Isosaari, P., Srivastava, V., Sillanpää, M., 2019. Ionic liquid-based water treatment technologies for organic pollutants: current status and future prospects of ionic liquid mediated technologies. *Sci. Total Environ.* 690, 604–619. <https://doi.org/10.1016/j.scitotenv.2019.06.421>.
- Jacso, P., 2005. The pros and cons of computing the h-index using Google Scholar. *Online Inf. Rev.* 29 (5), 588–596. <https://doi.org/10.1108/14684520810889718>.
- Jafarnejad, S., Jiang, S.C., 2019. Current technologies and future directions for treating petroleum refineries and petrochemical plants (PRPP) wastewaters. *J. Environ. Chem. Eng.* 7 (5), 103326 <https://doi.org/10.1016/j.jece.2019.103326>.
- James, A., 2022. Ceramic-microbial fuel cell (C-MFC) for waste water treatment: a mini review. *Environ. Res.* 210, 112963 <https://doi.org/10.1016/j.envres.2022.112963>.
- Karimidastanaei, Z., Klöve, B., Sadegh, M., Haghighi, A.T., 2021. Polar ice as an unconventional water resource: opportunities and challenges. *Water* 13 (22), 3220. <https://doi.org/10.3390/w13223220>.
- Karthigadevi, G., Manikandan, S., Karmegam, N., Subbaiya, R., Chozhavadhan, S., Ravindran, B., Chang, S.W., Awasthi, M.K., 2021. Chemico-nanotreatment methods for the removal of persistent organic pollutants and xenobiotics in water – a review. *Bioresour. Technol.* 324, 124678 <https://doi.org/10.1016/j.biortech.2021.124678>.
- Karaolia, P.A., 2019. Evaluation of the efficiency of the combination of a membrane bioreactor with selected advanced oxidation processes for the removal of antibiotic-related microcontaminants. PhD Thesis. University of Cyprus, Faculty of Engineering, Cyprus.
- Kataki, S., Chatterjee, S., Vairale, M.G., Sharma, S., Dwivedi, S.K., 2021. Concerns and strategies for wastewater treatment during COVID-19 pandemic to stop plausible transmission. *Resour. Conserv. Recycl.* 164, 105156 <https://doi.org/10.1016/j.resconrec.2020.105156>.
- Keeley, M., Koburger, A., Dolowitz, D.P., Medearis, D., Nickel, D., Shuster, W., 2013. Perspectives on the use of green infrastructure for stormwater management in Cleveland and Milwaukee. *Environ. Manag.* 51 (6), 1093–1108. <https://doi.org/10.1007/s00267-013-0032-x>.
- Khan, K.S., Kunz, R., Kleijnen, J., Antes, G., 2018. Five steps to conducting a systematic review. *J. R. Soc. Med.* 105 (12), 512–516. <https://doi.org/10.1177/0141076818802417>.
- Khan, P., Zhu, W., Huang, F., Gao, W., Khan, N.A., 2020. Micro-nanobubble technology and water-related application. *Water Supply* 20 (6), 2021–2035. <https://doi.org/10.2166/ws.2020.121>.
- Khan, M., Ahmad, A., Yuvaraj, M., Husain, S.S., Khan, D., 2023. A bibliometric analysis and visualization of global research on rural livelihood. *Global Knowledge, Memory and Communication*. <https://doi.org/10.1108/GKMC-10-2022-0231>.
- Kim, B.J., Jeong, S., Chung, J.B., 2021. Research trends in vulnerability studies from 2000 to 2019: findings from a bibliometric analysis. *Int. J. Disaster Risk Reduc.* 56, 102141 <https://doi.org/10.1016/j.ijdrr.2021.102141>.
- Kitchenham, B., Brereton, O.P., Budgen, D., Turner, M., Bailey, J., Linkman, S., 2009. Systematic literature reviews in software engineering—a systematic literature review. *Inf. Software Technol.* 51 (1), 7–15. <https://doi.org/10.1016/j.infsof.2008.09.009>.
- Kosek, K., Luczkiewicz, A., Fudala-Książek, S., Jankowska, K., Szopińska, M., Svahn, O., Tränckner, J., Kaiser, A., Langas, V., Björklund, E., 2020. Implementation of advanced micropollutants removal technologies in wastewater treatment plants (WWTPs) - examples and challenges based on selected EU countries. *Environ. Sci. Pol.* 112, 213–226. <https://doi.org/10.1016/j.envsci.2020.06.011>.
- Lasode, A., Rinn, E., Northrop, W.F., 2021. Surveying the applicability of energy recovery technologies for waste treatment: case study for anaerobic wastewater treatment in Minnesota. *J. Air Waste Manag. Assoc.* 71 (8), 974–988. <https://doi.org/10.1080/10962247.2021.1906353>.
- Leverett, D., 2014. In Response: what are the challenges and prospects? An industry perspective. *Environ. Toxicol. Chem.* 33 (11), 2407–2408. <https://doi.org/10.1002/etc.2714>.
- Li, J., See, K.F., Chi, J., 2019. Water resources and water pollution emissions in China's industrial sector: a green-biased technological progress analysis. *J. Clean. Prod.* 229, 1412–1426. <https://doi.org/10.1016/j.jclepro.2019.03.216>.
- Linnenluecke, M.K., Marrone, M., Singh, A.K., 2020. Conducting systematic literature reviews and bibliometric analyses. *Aust. J. Manag.* 45 (2), 175–194. <https://doi.org/10.1177/0312896219877678>.
- Liu, H., Kumar, V., Yadav, V., Guo, S., Sarsaiya, S., Binod, P., Kumar Awasthi, M., 2021a. Bioengineered biochar as smart candidate for resource recovery toward circular bio-economy: a review. *Bioengineered* 12 (2), 10269–10301. <https://doi.org/10.1080/21655979.2021.1993536>.
- Liu, H., Kumar, V., Yadav, V., Guo, S., Sarsaiya, S., Binod, P., et al., 2021b. Bioengineered biochar as smart candidate for resource recovery toward circular bio-economy: a review. *Bioengineered* 12 (2), 10269–10301. <https://doi.org/10.1080/21655979.2021.1993536>.
- Lyu, Y., Liu, Y., Guo, Y., Tian, J., Chen, L., 2021. Managing water sustainability in textile industry through adaptive multiple stakeholder collaboration. *Water Res.* 205, 117655 <https://doi.org/10.1016/j.watres.2021.117655>.
- Madikizela, L.M., Ncube, S., Chimuka, L., 2020. Analysis, occurrence and removal of pharmaceuticals in African water resources: a current status. *J. Environ. Manag.* 253, 109741 <https://doi.org/10.1016/j.jenvman.2019.109741>.
- Magro, C., Mateus, E.P., Paz-Garcia, J.M., Ribeiro, A.B., 2020. Emerging organic contaminants in wastewater: understanding electrochemical reactors for triclosan and its by-products degradation. *Chemosphere* 247, 125758. <https://doi.org/10.1016/j.chemosphere.2019.125758>.
- Magwaza, S.T., Magwaza, L.S., Odindo, A.O., Mditshwa, A., 2020. Hydroponic technology as decentralised system for domestic wastewater treatment and vegetable production in urban agriculture: a review. *Sci. Total Environ.* 698, 134154 <https://doi.org/10.1016/j.scitotenv.2019.134154>.
- Man, Y., Han, Y., Wang, Y., Li, J., Chen, L., Qian, Y., Hong, M., 2018. Woods to goods: water consumption analysis for papermaking industry in China. *J. Clean. Prod.* 195, 1377–1388. <https://doi.org/10.1016/j.jclepro.2018.06.026>.
- Martin Ruel, S., Choubert, J.-M., Budzinski, H., Mège, C., Esperanza, M., Coquery, M., 2012. Occurrence and fate of relevant substances in wastewater treatment plants regarding Water Framework Directive and future legislations. *Water Sci. Technol.* 65 (7), 1179–1189. <https://doi.org/10.2166/wst.2012.943>.
- McCance, W., Jones, O.A.H., Edwards, M., Surapaneni, A., Chadalavada, S., Currell, M., 2018. Contaminants of Emerging Concern as novel groundwater tracers for delineating wastewater impacts in urban and peri-urban areas. *Water Res.* 146, 118–133. <https://doi.org/10.1016/j.watres.2018.09.013>.
- McCance, W., Jones, O.A.H., Cendón, D.I., Edwards, M., Surapaneni, A., Chadalavada, S., Currell, M., 2021. Decoupling wastewater impacts from hydrogeochemical trends in impacted groundwater resources. *Sci. Total Environ.* 774, 145781 <https://doi.org/10.1016/j.scitotenv.2021.145781>.
- McQuaid, S., Kooijman, E.D., Rhodes, M.L., Cannon, S.M., 2021. Innovating with nature: factors influencing the success of nature-based enterprises. *Sustainability* 13 (22), 12488. <https://doi.org/10.3390/su132212488>.
- Mekuria, D.M., Kassegne, A.B., Asfaw, S.L., 2021. Assessing pollution profiles along Little Akaki River receiving municipal and industrial wastewaters, Central Ethiopia: implications for environmental and public health safety. *Heliyon* 7 (7), e07526. <https://doi.org/10.1016/j.heliyon.2021.e07526>.
- Moggio, L., de Sire, A., Marotta, N., Demeco, A., Ammendolia, A., 2022. Vibration therapy role in neurological diseases rehabilitation: an umbrella review of systematic reviews. *Disabil. Rehabil.* 44 (20), 5741–5749. <https://doi.org/10.1080/09638288.2021.1946175>.
- Moher, D., Liberati, A., Tetzlaff, J., Altman, D.G., Prisma Group*, T., 2009. Preferred reporting items for systematic reviews and meta-analyses: the PRISMA statement. *Ann. Intern. Med.* 151 (4), 264–269. <https://doi.org/10.3736/jcm20090918>.
- Mongeon, P., Paul-Hus, A., 2016. The journal coverage of Web of Science and Scopus: a comparative analysis. *Scientometrics* 106, 213–228. <https://doi.org/10.1007/s11192-015-1765-5>.
- Morris, L., Colombo, V., Hassell, K., Kellar, C., Leahy, P., Long, S.M., Myers, J.H., Pettigrove, V., 2017. Municipal wastewater effluent licensing: a global perspective and recommendations for best practice. *Sci. Total Environ.* 580, 1327–1339.
- Moussavi, S., Thompson, M., Li, S., Dvorak, B., 2021. Assessment of small mechanical wastewater treatment plants: relative life cycle environmental impacts of construction and operations. *J. Environ. Manag.* 292, 112802 <https://doi.org/10.1016/j.jenvman.2021.112802>.

- Mullins, D., Jones, E., Glavin, M., Coburn, D., Hannon, L., Clifford, E., 2018. A novel image processing-based system for turbidity measurement in domestic and industrial wastewater. *Water Sci. Technol.* 77 (5), 1469–1482. <https://doi.org/10.2166/wst.2018.030>.
- Neale, P.A., Jämting, A.K., Escher, B.I., Herrmann, J., 2013. A review of the detection, fate and effects of engineered nanomaterials in wastewater treatment plants. *Water Sci. Technol.* 68 (7), 1440–1453. <https://doi.org/10.2166/wst.2013.388>.
- Norgate, T., Haque, N., 2010. Energy and greenhouse gas impacts of mining and mineral processing operations. *J. Clean. Prod.* 18 (3), 266–274. <https://doi.org/10.1016/j.jclepro.2009.09.020>.
- Nudurupati, S.S., Budhwar, P., Pappu, R.P., Chowdhury, S., Kondala, M., Chakraborty, A., Ghosh, S.K., 2022. Transforming sustainability of Indian small and medium-sized enterprises through circular economy adoption. *J. Bus. Res.* 149, 250–269. <https://doi.org/10.1016/j.jbusres.2022.05.036>.
- Nunes, O.C., 2021. The challenge of removing waste from wastewater: let technology use nature. *Microb. Biotechnol.* 14 (1), 63–67. <https://doi.org/10.1111/1751-7915.13711>.
- Oladinrin, O.T., Arif, M., Rana, M.Q., Gyoh, L., 2023. Interrelations between construction ethics and innovation: a bibliometric analysis using VOSviewer. *Construct. Innovat.* 23 (3), 505–523. <https://doi.org/10.1108/CI-07-2021-0130>.
- Ortúzar, M., Esterhuizen, M., Olicón-Hernández, D.R., González-López, J., Aranda, E., 2022. Pharmaceutical pollution in aquatic environments: a concise review of environmental impacts and bioremediation systems. *Front. Microbiol.* 13, 869332. <https://doi.org/10.3389/fmicb.2022.869332>.
- Ozor, P.A., Mbohwa, C., 2018. Characterization of sewage treatment plant in an emerging territory. In: *Proceedings of the World Congress on Engineering*, vol. 2. https://www.iaeng.org/publication/WCE2018/WCE2018_pp712-715.pdf.
- O'Mara Eves, A., Thomas, J., McNaught, J., Miwa, M., Ananiadou, S., 2015. Using text mining for study identification in systematic reviews: a systematic review of current approaches. *Syst. Rev.* 4 (1), 1–22. <https://doi.org/10.1186/2046-4053-4-5>.
- Page, M.J., McKenzie, J.E., Bossuyt, P.M., Boutron, I., Hoffmann, T.C., Mulrow, C.D., Moher, D., 2021. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. *International journal of surgery* 88, 105906.
- Pandey, A.K., Reji Kumar, R., B.K. Laghari, I.A., Samykano, M., Kothari, R., Abusorrah, A.M., Sharma, K., Tyagi, V.V., 2021. Utilization of solar energy for wastewater treatment: challenges and progressive research trends. *J. Environ. Manag.* 297, 113300. <https://doi.org/10.1016/j.jenvman.2021.113300>.
- Patterson, J.W., 1985. *Industrial Wastewater Treatment Technology*. U.S. Department of Energy, Office of Scientific and Technical Information. <https://www.osti.gov/biblio/7253209>.
- Pau, C., Serra, T., Colomer, J., Casamitjana, X., Sala, L., Kampf, R., 2013. Filtering capacity of *Daphnia magna* on sludge particles in treated wastewater. *Water Res.* 47 (1), 181–186. <https://doi.org/10.1016/j.watres.2012.09.047>.
- Perez, J.P.H., Folems, K., Leus, K., Vanhaecke, F., Van Der Voort, P., Du Laing, G., 2019. Progress in hydrometallurgical technologies to recover critical raw materials and precious metals from low-concentrated streams. *Resour. Conserv. Recycl.* 142, 177–188. <https://doi.org/10.1016/j.resconrec.2018.11.029>.
- Petcharat, T., Kongprajug, A., Chyerochana, N., Sangkaew, W., Mongkolsuk, S., Sirikanthana, K., 2020. Assessing human-specific CrAssphage recovery after acidification-filtration concentrating method in environmental water. *Water Environ. Res.* 92 (1), 35–41. <https://doi.org/10.1002/wer.1209>.
- Plósz, B.G., Benedetti, L., Daigger, G.T., Langford, K.H., Larsen, H.F., Monteith, H., Vanrolleghem, P.A., 2013. Modelling micro-pollutant fate in wastewater collection and treatment systems: status and challenges. *Water Sci. Technol.* 67 (1), 1–15. <https://doi.org/10.2166/wst.2012.562>.
- Priyadarshini, M., Das, I., Ghangrekar, M.M., Blaney, L., 2022. Advanced oxidation processes: performance, advantages, and scale-up of emerging technologies. *J. Environ. Manag.* 316, 115295. <https://doi.org/10.1016/j.jenvman.2022.115295>.
- Qu, J., Wang, H., Wang, K., Yu, G., Ke, B., Yu, H.-Q., Ren, H., Zheng, X., Li, J., Li, W.-W., Gao, S., Gong, H., 2019. Municipal wastewater treatment in China: development history and future perspectives. *Front. Environ. Sci. Eng.* 13 (6), 88. <https://doi.org/10.1007/s11783-019-1172-x>.
- R van Dinter, R., Catal, C., Tekinerdogan, B., 2021. A Multi-Channel Convolutional Neural Network approach to automate the citation screening process. *Appl. Soft Comput.* 112, 107765. <https://doi.org/10.1016/j.asoc.2021.107765>.
- Rahman, T.U., Roy, H., Islam, M.R., Tahmid, M., Fariha, A., Mazumder, A., Tasnim, N., Pervez, M.N., Cai, Y., Naddeo, V., Islam, M.S., 2023. The advancement in membrane bioreactor (MBR) technology toward sustainable industrial wastewater management. *Membranes* 13 (2), 181. <https://doi.org/10.3390/membranes13020181>.
- Rana, A., Sudhaik, A., Raizada, P., Nguyen, V.-H., Xia, C., Parwaz Khan, A.A., Thakur, S., Nguyen-Tri, P., Nguyen, C.C., Kim, S.Y., Le, Q.V., Singh, P., 2022. Graphitic carbon nitride based immobilized and non-immobilized floating photocatalysts for environmental remediation. *Chemosphere* 297, 134229. <https://doi.org/10.1016/j.chemosphere.2022.134229>.
- Ray, R., Kapadia, G., Sambasivam, S., 2015. Liquid waste management in E&P-cairn India case study. In: *Abu Dhabi International Petroleum Exhibition and Conference. OnePetro*. <https://doi.org/10.2118/177920-MS>. November.
- Reichwaldt, E.S., Ho, W.Y., Zhou, W., Ghadouani, A., 2017. Sterols indicate water quality and wastewater treatment efficiency. *Water Res.* 108, 401–411. <https://doi.org/10.1016/j.watres.2016.11.029>.
- Rice, E.W., Bridgewater, L., American Public Health Association (Eds.), 2012. *Standard Methods for the Examination of Water and Wastewater*, vol. 10. American public health association, Washington, DC. <http://yabesh.ir/wp-content/uploads/2018/02/Standard-Methods-23rd-Perv.pdf>.
- Rippin, Sharma, A.K., Beniwal, V., 2022. Biosynthesis and medicinal applications of proanthocyanidins: a recent update. *Biocatal. Agric. Biotechnol.* 102500. <https://doi.org/10.1016/j.bcab.2022.102500>.
- Rippin, Sharma, A.K., Beniwal, V., 2023a. Reconnoitring the antioxidant and antibacterial potential of different fruits after tannin acyl hydrolase mediated biotransformation. *Biotechnol. Appl. Biochem.* <https://doi.org/10.1002/bab.2461>.
- Rippin, Beniwal, V., Sharma, A., Singh, B.J., Ramniwas, S., Sak, K., Kumar, S., Sharma, A.K., 2023b. Ginnalin A and hamamelitannin: the unique gallotannins with promising anti-carcinogenic potential. *Explor. Target. Antitumor Ther.* 4 (2), 208. <https://doi.org/10.37349/etat.2023.00129>.
- Rodrigues de Assis, L., Calijuri, M.L., Assemany, P.P., Silva, T.A., Teixeira, J.S., 2020. Innovative hybrid system for wastewater treatment: high-rate algal ponds for effluent treatment and biofilm reactor for biomass production and harvesting. *J. Environ. Manag.* 274, 111183. <https://doi.org/10.1016/j.jenvman.2020.111183>.
- Ruane, J., Sonnino, A., Steduto, P., Deane, C., 2008. Coping with Water Scarcity: what Role for Biotechnologies. *Land and Water Discussion Paper*, 7. https://www.researchgate.net/profile/Andrea-Sonnino2/publication/236848232_Coping_with_Water_Scarcity_What_Role_for_Biotechnologies/links/568bd20208ae16c4149bf7e/Coping-with-Water-Scarcity-What-Role-for-Biotechnologies.pdf, 19th Aug 2023.
- Saheb, T., Izadi, L., 2019. Paradigm of IoT big data analytics in the healthcare industry: a review of scientific literature and mapping of research trends. *Telematics Inf.* 41, 70–85. <https://doi.org/10.1016/j.tele.2019.03.005>.
- Sancho, I., Lopez-Palau, S., Arespacochaga, N., Cortina, J.L., 2019. New concepts on carbon redirection in wastewater treatment plants: a review. *Sci. Total Environ.* 647, 1373–1384. <https://doi.org/10.1016/j.scitotenv.2018.08.070>.
- Santos, E., Albuquerque, A., Lisboa, I., Murray, P., Ermis, H., 2022. Economic assessment of energy consumption in wastewater treatment plants: applicability of alternative nature-based technologies in Portugal. *Water (Switzerland)* 14 (13), 2042. <https://doi.org/10.3390/w14132042>.
- Saravanan, A., Senthil Kumar, P., Jeevanantham, S., Karishma, S., Tajsabreen, B., Yaashika, P.R., Reshma, B., 2021. Effective water/wastewater treatment methodologies for toxic pollutants removal: processes and applications towards sustainable development. *Chemosphere* 280, 130595. <https://doi.org/10.1016/j.chemosphere.2021.130595>.
- Schäfer, M., Hobus, I., Schmitt, T.G., 2017. Energetic flexibility on wastewater treatment plants. *Water Sci. Technol.* 76 (5), 1225–1233. <https://doi.org/10.2166/wst.2017.308>.
- Shea, B.J., Bouter, L.M., Peterson, J., Boers, M., Andersson, N., Ortiz, Z., Ramsay, T., Bai, A., Shukla, V.K., Grimshaw, J.M., 2007. External validation of a measurement tool to assess systematic reviews (AMSTAR). *PLoS One* 2 (12), e1350. <https://doi.org/10.1371/journal.pone.0001350>.
- Shen, C., 2018. A transdisciplinary review of deep learning research and its relevance for water resources scientists. *Water Resour. Res.* 54 (11), 8558–8593.
- Sheth, R.P., Ranawat, N.S., Chakraborty, A., Mishra, R.P., Khandelwal, M., 2023. The lithium-ion battery recycling process from a circular economy perspective—a review and future directions. *Energies* 16 (7), 3228. <https://doi.org/10.3390/en16073228>.
- Silombela, T., Mutingi, M., Chakraborty, A., 2018. Impact of quality management tools and techniques: case of Namibian municipal water distributors. *J. Qual. Mainten. Eng.* 24 (1), 2–21. <https://doi.org/10.1108/JQME-05-2016-0020>.
- Sindane, T.J., Modley, S.L.A., 2022. The impacts of poor water quality on the residential areas of Emfuleni local municipality: a case study of perceptions in the Rietspruit River catchment in South Africa. *Urban Water J.* 1–11. <https://doi.org/10.1080/1573062X.2022.2030763>.
- Singh, R.P., Singh, P.K., Gupta, R., Singh, R.L., 2019. Treatment and recycling of wastewater from textile industry. *Advances in biological treatment of industrial waste water and their recycling for a sustainable future* 225–266. https://doi.org/10.1007/978-981-13-1468-1_8.
- Skoulidakis, Ch., Ganoulis, J., 2012. *Assessment of Environmental Values for Integrated Water Resources Management*. In: Katsifarakis, K.L., et al. (Eds.), *Inter-national Conference of Protection and Restoration of the Environment*, Thessaloniki, XI, pp. 253–261.
- Sodhi, H.S., Singh, D., Singh, B., 2013. Lean and six sigma: a combined approach for waste management in Indian SME's. *International Journal of Latest Technology in Engineering, Management & Applied Science* 4 (4), 7–12. (Accessed 30 August 2023).
- Sodhi, H.S., Singh, D., Singh, B.J., 2020a. A conceptual examination of Lean, Six Sigma and Lean Six Sigma models for managing waste in manufacturing SMEs. *World J. Sci. Technol. Sustain. Dev.* 17 (1), 20–32. <https://doi.org/10.1108/WJSTSD-10-2019-0073>.
- Sodhi, H.S., Singh, D., Singh, B.J., 2020b. An investigation of barriers to waste management techniques implemented in Indian manufacturing industries using analytical hierarchy process. *World J. Sci. Technol. Sustain. Dev.* 17 (1), 58–70. <https://doi.org/10.1108/WJSTSD-09-2019-0069>.
- Sodhi, H.S., Singh, D., Singh, B.J., 2020c. SWOT analysis of waste management techniques quantitatively. *Int. J. Adv. Oper. Manag.* 12 (2), 103–121. <https://doi.org/10.1504/IJAOM.2020.108260>.
- Sodhi, H.S., Singh, D., Singh, B.J., 2022. A qualitative and quantitative comparison of various waste management techniques. *Int. J. Bus. Excel.* 28 (3), 350–363. <https://doi.org/10.1504/IJBEX.2022.126923>.
- Somlyódy, L., Patziger, M., 2012. Urban wastewater development in central and eastern europe. *Water Sci. Technol.* 66 (5), 1081–1087. <https://doi.org/10.2166/wst.2012.289>.
- Spina, F., Gea, M., Bicchì, C., Cordero, C., Schilirò, T., Varese, G.C., 2020. Ecofriendly laccases treatment to challenge micropollutants issue in municipal wastewaters. *Environ. Pollut.* 257, 113579. <https://doi.org/10.1016/j.envpol.2019.113579>.

- Stamatelatou, K., Tsagarakis, K.P., 2015. Sewage Treatment Plants: Economic Evaluation of Innovative Technologies for Energy Efficiency. IWA Publishing. <https://library.oapen.org/handle/20.500.12657/43789>, 31st Aug 2023.
- Subhiksha, V., Kokilavani, S., Sudheer Khan, S., 2022. Recent advances in degradation of organic pollutant in aqueous solutions using bismuth based photocatalysts: a review. *Chemosphere* 290, 133228. <https://doi.org/10.1016/j.chemosphere.2021.133228>.
- Tallon, P.P., 2013. Corporate governance of big data: perspectives on value, risk, and cost. *Computer* 46 (6), 32–38. <https://doi.org/10.1109/MC.2013.155>.
- Tamala, J.K., Maramag, E.I., Simeon, K.A., Ignacio, J.J., 2022. A bibliometric analysis of sustainable oil and gas production research using VOSviewer. *Clean. Eng. Technol.* 7, 100437. <https://doi.org/10.1016/j.clet.2022.100437>.
- Tlili, A., Corcoll, N., Arrhenius, Å., Backhaus, T., Hollender, J., Creusot, N., Wagner, B., Behra, R., 2020. Tolerance patterns in stream biofilms link complex chemical pollution to ecological impacts. *Environ. Sci. Technol.* 54 (17), 10745–10753. <https://doi.org/10.1021/acs.est.0c02975>.
- Tong, Y., Huang, Z., Janssen, A.B.G., Wishart, M., He, W., Wang, X., Zhao, Y., 2022. Influence of social and environmental drivers on nutrient concentrations and ratios in lakes: a comparison between China and Europe. *Water Res.* 227, 119347. <https://doi.org/10.1016/j.watres.2022.119347>.
- Trianni, A., Negri, M., Cagno, E., 2021. What factors affect the selection of industrial wastewater treatment configuration? *J. Environ. Manag.* 285, 112099. <https://doi.org/10.1016/j.jenvman.2021.112099>. Get rights and content.
- United Nations Environment Programme, 2017. Global Environment Outlook 6. Retrieved from. <https://www.unenvironment.org/resources/global-environment-outlook-6>. (Accessed 12 November 2022).
- Verdaguer, M., Aulinas, M., Clara, N., Poch, M., 2010. Integrating swarm intelligence in a Multi-Agent System for wastewater discharges management. In: 2010 5th IEEE International Conference Intelligent Systems. IEEE, pp. 444–449. <https://doi.org/10.1109/IS.2010.5548341>. July.
- Verstraete, W., Yanuka-Golub, K., Driesen, N., De Vrieze, J., 2022. Engineering microbial technologies for environmental sustainability: choices to make. *Microb. Biotechnol.* 15 (1), 215–227. <https://doi.org/10.1111/1751-7915.13986>.
- Vrzala, M., Goño, M., Goño, R., Kotulla, M., Wzorek, M., Leonowicz, Z., 2022. Distributed generation power systems in wastewater management. *Energies* 15 (17), 6283. <https://doi.org/10.3390/en15176283>.
- Vystavna, Y., Frkova, Z., Celle-Jeanton, H., Diadin, D., Huneau, F., Steinmann, M., Loup, C., 2018. Priority substances and emerging pollutants in urban rivers in Ukraine: occurrence, fluxes and loading to transboundary European Union watersheds. *Sci. Total Environ.* 637, 1358–1362. <https://doi.org/10.1016/j.scitotenv.2018.05.095>.
- Wang, Z., Zhao, Y., Wang, B., 2018. A bibliometric analysis of climate change adaptation based on massive research literature data. *J. Clean. Prod.* 199, 1072–1082. <https://doi.org/10.1016/j.jclepro.2018.06.183>.
- Wang, B., Cui, H., Liu, H., Wan, Y., 2019. Derivatization for nontargeted screening of acids in oilfield refinery wastewater: identification and behaviors of recalcitrant chlorinated naphthenic acids. *Environ. Sci. Technol.* 53 (2), 1022–1030. <https://doi.org/10.1021/acs.est.8b05310>.
- Wang, Y., Fan, L., Khan, S.J., Roddick, F.A., 2020. Fugacity modelling of the fate of micropollutants in aqueous systems — uncertainty and sensitivity issues. *Sci. Total Environ.* 699, 134249. <https://doi.org/10.1016/j.scitotenv.2019.134249>.
- Warner, N.R., Christie, C.A., Jackson, R.B., Vengosh, A., 2013. Impacts of shale gas wastewater disposal on water quality in Western Pennsylvania. *Environ. Sci. Technol.* 47 (20), 11849–11857. <https://doi.org/10.1021/es402165b>.
- Wawale, S.G., Shabaz, M., Mehbodniya, A., Soni, M., Deb, N., Elashiri, M.A., Naved, M., 2022. Biomedical waste management using IoT tracked and fuzzy classified integrated technique. *Human-centric Computing and Information Sciences* 12 (32). <http://hccisj.com/data/file/article/2022070002/12-32.pdf>, 30th Aug 2023.
- Wicklein, E., Batstone, D.J., Ducoste, J., Laurent, J., Griborio, A., Wicks, J., et al., 2016. Good modelling practice in applying computational fluid dynamics for WWTP modelling. *Water Sci. Technol.* 73 (5), 969–982. <https://doi.org/10.2166/wst.2015.565>.
- Wohlfart, O., Wagner, I., 2023. Teachers' role in digitalizing education: an umbrella review. *Educ. Technol. Res. Dev.* 71 (2), 339–365. <https://doi.org/10.1007/s11423-022-10166-0>.
- Wolf, Y., Oster, S., Shuliakovich, A., Brückner, I., Dolny, R., Linnemann, V., Pinnekamp, J., Hollert, H., Schiwly, S., 2022. Improvement of wastewater and water quality via a full-scale ozonation plant? – a comprehensive analysis of the endocrine potential using effect-based methods. *Sci. Total Environ.* 803, 149756. <https://doi.org/10.1016/j.scitotenv.2021.149756>.
- Xi, H., Zhou, X., Arslan, M., Luo, Z., Wei, J., Wu, Z., Gamal El-Din, M., 2022. Heterotrophic nitrification and aerobic denitrification process: promising but a long way to go in the wastewater treatment. *Sci. Total Environ.* 805, 150212. <https://doi.org/10.1016/j.scitotenv.2021.150212>.
- Xu, A., Wu, Y.H., Chen, Z., Wu, G., Wu, Q., Ling, F., Huang, W.E., Hu, H.Y., 2020. Towards the new era of wastewater treatment of China: development history, current status, and future directions. *Water Cycle* 1, 80–87. <https://doi.org/10.1016/j.watcyc.2020.06.004>.
- Yang, Y., Shi, X., Ballent, W., Mayer, B.K., 2017. Biological phosphorus recovery: review of current progress and future needs: yang et al. *Water Environ. Res.* 89 (12), 2122–2135. <https://doi.org/10.2175/106143017X15054988926424>.
- Yang, Z., Qian, J., Shan, C., Li, H., Yin, Y., Pan, B., 2021. Toward selective oxidation of contaminants in aqueous systems. *Environ. Sci. Technol.* 55 (21), 14494–14514. <https://doi.org/10.1021/acs.est.1c05862>.
- Ye, Z., Yang, J., Zhong, N., Tu, X., Jia, J., Wang, J., 2020. Tackling environmental challenges in pollution controls using artificial intelligence: a review. *Sci. Total Environ.* 699, 134279. <https://doi.org/10.1016/j.scitotenv.2019.134279>.
- Yu, S., Tang, H., Zhang, D., Wang, S., Qiu, M., Song, G., Fu, D., Hu, B., Wang, X., 2022. MXenes as emerging nanomaterials in water purification and environmental remediation. *Sci. Total Environ.* 811, 152280. <https://doi.org/10.1016/j.scitotenv.2021.152280>.
- Zakaria, Y., Michael, K., 2017. An integrated cloud-based wireless sensor network for monitoring industrial wastewater discharged into water sources. *Wirel. Sens. Netw.* 9 (8), 290. <https://doi.org/10.4236/wsn.2017.98016>.
- Zan, F., Iqbal, A., Lu, X., Wu, X., Chen, G., 2022. “Food waste-wastewater-energy/resource” nexus: integrating food waste management with wastewater treatment towards urban sustainability. *Water Res.* 211, 118089. <https://doi.org/10.1016/j.watres.2022.118089>.
- Zhang, Q.H., Yang, W.N., Ngo, H.H., Guo, W.S., Jin, P.K., Dzakpasu, M., Yang, S.J., Wang, Q., Wang, X.C., Ao, D., 2016. Current status of urban wastewater treatment plants in China. *Environ. Int.* 92–93, 11–22. <https://doi.org/10.1016/j.envint.2016.03.024>.
- Zou, Y., Li, Q., Tao, T., Ye, P., Zhang, P., Liu, Y., 2021. Fe-Mn binary oxides activated aluminosilicate mineral and its Tl (I) removal by oxidation, precipitation and adsorption in aqueous. *J. Solid State Chem.* 303, 122383. <https://doi.org/10.1016/j.jssc.2021.122383>.
- Zulkifli, S.N., Rahim, H.A., Lau, W.J., 2018. Detection of contaminants in water supply: a review on state-of-the-art monitoring technologies and their applications. *Sensor. Actuators. B Chem.* 255, 2657–2689. <https://doi.org/10.1016/j.snb.2017.09.078>.
- Zupic I, Čater T. Bibliometric methods in management and organization. *Organ. Res. Methods*, 18(3), 429–472. <https://doi.org/10.1177/1094428120969905>.